What Do Group-Conditional Coverage Metrics Measure? Identification, Inference and Auditing for Conformal Evaluation

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Abstract

Conformal methods for group-conditional or class-conditional coverage are compared using metrics computed from the empirical coverage of each group on a test set, such as the average coverage gap, the fraction of under-covered groups and the worst-group coverage. When groups are many and small, much of what these metrics report is binomial test noise. We separate the method's expected coverage, the coverage realized by one calibration draw, and the test noise, and study what the data identify about the realized coverage. We give closed-form noise floors, the values that a method with exact group coverage would show. The fraction of under-covered groups and the worst group are not identified when groups have few test points; we give their sharp identified sets. The root-mean-square coverage gap is point-identified. For it we give a debiased estimator, an asymptotically valid interval whose length is rate-optimal up to a logarithmic factor, finite-sample upper bounds, and an asymptotic paired comparison of methods. To flag individual groups, BH on exact binomial p-values controls the false discovery rate in finite samples, and its power for a group is at most that of a single exact test. In a pre-registered evaluation on held-out real data (7 datasets, 7 methods, 200 splits), test noise is a median 41% of the naive mean-squared coverage gap, ranging from 5% to nearly all of it, and a method with exact group coverage would show a median 62% of the reported coverage gap. The finite-sample bounds are too conservative to certify this overstatement at these test-set sizes. Post hoc, the method with the largest gap (Mondrian) mostly over-covers: its estimated shortfall is below standard split's in 24 of 26 settings.

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1. Introduction

Split conformal prediction guarantees marginal coverage (Vovk et al., 2005; Lei et al., 2018). Much recent work targets coverage conditional on a group or a class (Vovk, 2012; Ding et al., 2023; Gibbs et al., 2025), and new methods are compared on metrics that summarize per-group empirical coverage on a test set: the mean absolute coverage gap (CovGap), the fraction of groups whose coverage falls below a cut, and the worst-group coverage.

These metrics are plug-in functionals of the empirical coverages $\hat{c}_g = h_g/m_g$, where m_g test points fall in group g. When there are many groups and few test points per group, which is exactly the regime that motivates group-conditional methods, $\hat{c}_g$ is dominated by binomial noise. A method with *exact* group-conditional coverage 0.9 evaluated on groups with five test points each shows a CovGap of 0.118 and 41% of its groups "below 0.85" (Proposition 3.2). The size of a reported gap therefore depends on the test counts as much as on the method.

What we do. Section 2 separates three layers of randomness in group-coverage evaluation and takes as target the coverage realized by the deployed calibration draw. Section 3 derives the exact bias of the naive metrics and their values under exact validity, including for Mondrian conformal prediction. Section 4 shows that threshold metrics (fraction below a cut, quantiles, worst group) are not identified when groups have few test points, with sharp identified sets for $m_g \leq 2$ and moment-LP bounds in general; CovGap is only partially identified, but it is bounded by the RMS gap, which is point-identified. Section 5 gives an interval for the RMS gap that needs no model for how the c_g arise. No valid interval can be shorter than order $G^{-1/4}$ near zero, and ours attains this rate up to constants and a $(\log m_{\max})^{1/4}$ factor. Its upper end can under-cover, so we add finite-sample upper bounds, the tighter of which uses the exact binomial moment generating function, and compute how many groups they need to certify that a reported CovGap overstates the realized one. The same section gives an asymptotic paired interval for comparing two methods on the same test points. Section 6 shows that BH on exact binomial p-values controls the false discovery rate of a group

audit in finite samples, which popular alternatives do not, and that its power for a group is capped by that of a single exact test.

What we find. The studies in Section 7 were pre-registered. On synthetic data with known c_g, our estimators beat the comparators in 87–90% of cells, and the audit controls the FDR where popular alternatives reach 0.19–0.62. The asymptotic RMS interval misses above its upper end when groups are small and coverage is high, because the debiased terms are then right-skewed; the finite-sample bounds were exceeded in at most 3 of 672,000 replications. The paired interval fails in the same way when one method covers almost always in small groups; a rule that excludes those cases and requires at least 200 groups held on a fresh grid, but with many small groups no comparison we tried is both valid and informative. On held-out real data, test noise is a median 41% of the naive mean-squared gap, and nearly all of it in one dataset. The naive metrics mostly preserve the ranking of methods. Mondrian, which has the largest gap, mostly over-covers: its estimated shortfall is below standard split's in 24 of 26 runs (post hoc). None of our finite-sample bounds certifies the overstatement at these test-set sizes, and whether any valid bound can is open.

The statistical tools we use (binomial deconvolution, moment problems, debiased quadratic metrics, multiple testing) are known, and Section 8 lists what we take from where. What is new is their use for conformal evaluation, the conformal-specific results (noise floors, identified sets for coverage metrics, the length bound for heterogeneous m_g, the finite-sample certificate), and the measured size of the correction on held-out data.

2. Setup: three layers of group coverage

A conformal method is calibrated on D_{cal} at level α; write $\tau = 1 - \alpha$. Groups $g = 1, \dots, G$ form a pre-specified partition of the feature space, or the labels in class-conditional prediction. The *realized coverage* of group g is

$$c_g = \mathbb{P}\big(Y \in C(X) \mid X \in g,\ D_{\text{cal}}\big) \in [0, 1]. \qquad (1)$$

Test points are i.i.d. from the population and independent of D_{cal}; m_g of them fall in group g and h_g of those are covered, so $\hat{c}_g = h_g/m_g$.

Assumption 2.1 (Test layer)**.** Conditionally on D_{cal} and on the group labels of the test points, $h_g \sim \text{Bin}(m_g, c_g)$ independently over g.

Assumption 2.1 says nothing about how the c_g arise: they may be dependent through the shared calibration set, as for most conformal methods. Our targets are coverages on new draws from the population, so Assumption 2.1 asks that test records be independent within a group. A random split makes the test set independent of D_{cal} but cannot remove dependence among the records, such as flights on the same stormy day. Coverage on the finite data set itself is a different target with almost no test noise, since the test set is most of the data; it suits statements about one fixed benchmark split, not statements about the methods. We measured the dependence: in the airline and Zurich data, coverage varies across days within a group only 2–4% more than random reallocation of the days produces (Appendix B.2). The binomial variance, and with it the floor and the debiasing, is then understated by about that much. We did not check other forms of clustering (multi-day events, routes, households in ACS), so on real data the finite-sample guarantees hold only approximately. For ACS, targets refer to the unweighted sample, not to the survey-weighted population.

Three layers. (a) The method's expected group coverage $\mathbb{E}[c_g]$, the object of conformal guarantees; (b) the realized coverage c_g of the deployed calibration draw, which is what a practitioner gets; (c) the test noise $\hat{c}_g - c_g$. Published group metrics apply a functional to the $\hat{c}_g$. We take as targets the same functionals applied to the c_g (layer b), conditionally on D_{cal} and $m = (m_g)_g$. Layers (a) and (b) answer different questions: how unevenly the deployed calibration covers the groups (b), and whether the method treats some groups worse in expectation (a). A guarantee and a gap can disagree: Mondrian prediction satisfies $\mathbb{E}[c_g] \geq \tau$ in every group, yet has the largest gap at both layers, mostly from systematic over-coverage of groups with few calibration points, which $|c_g - \tau|$ penalizes too, while its one-sided shortfall is the smallest (Appendix B.8). Averaging a layer-(b) metric over calibration draws does not answer (a): for the MSE it estimates $\text{mean}_g[(\mathbb{E}c_g - \tau)^2 + \text{Var}c_g]$. With K disjoint calibration–test blocks, $\sum_{k \neq l}(\hat{c}_g^{(k)} - \tau)(\hat{c}_g^{(l)} - \tau)/(K(K-1))$ is unbiased for $(\mathbb{E}c_g - \tau)^2$; average it over groups and use a jackknife over blocks (Appendix A). It targets the method at the blocks' calibration size, at most n/K for a data set of size n, which matters for methods such as Mondrian. Overlapping splits have no unbiased variance estimator (Bengio & Grandvalet, 2004), and resampled tests are approximate (Dietterich, 1998; Nadeau & Bengio, 2003).

Target functionals. With $\mathcal{G}_1 = \{g : m_g \geq 1\}$ and $\mathcal{G}_2 = \{g : m_g \geq 2\}$,

$$\text{CovGap} = \operatorname*{mean}_{\mathcal{G}_1} |c_g - \tau|, \quad \text{SF} = \operatorname*{mean}_{\mathcal{G}_1} (\tau - c_g)_+,$$

$$\text{MSE} = \operatorname*{mean}_{\mathcal{G}_2} (c_g - \tau)^2, \quad \text{RMS} = \sqrt{\text{MSE}},$$

$$\text{FracBelow}(t) = \operatorname*{mean}_{\mathcal{G}_1} \mathbf{1}\{c_g < t\}, \quad \text{Worst} = \min_{\mathcal{G}_1} c_g.$$

SF is the mean shortfall. These are *compound* targets: functionals of the fixed vector c, not of a superpopulation. The naive estimator of each is the same functional of $\hat{c}$.

Random-effects model (identification only). Statements about identification use a random-effects (RE) model in which, within a stratum s of groups with similar m_g, the c_g are i.i.d. from a law ν_s independent of the m_g. The compound law $\tilde{\nu}_s = G_s^{-1} \sum_{g \in s} \delta_{c_g}$ then satisfies $W_1(\tilde{\nu}_s, \nu_s) \to 0$ a.s. Inference for the RMS gap (Section 5) and the auditing procedure (Section 6) do *not* use this model.

3. What the naive metrics measure

Proposition 3.1 (Naive bias)**.** *Under Assumption 2.1: (a)* $\mathbb{E}[(\hat{c}_g - \tau)^2 \mid c] = (c_g - \tau)^2 + c_g(1 - c_g)/m_g$*, so the naive MSE overstates* MSE *by* $\mathrm{mean}_{\mathcal{G}_2} c_g(1 - c_g)/m_g$ *in expectation; (b)* $\mathbb{E}[|\hat{c}_g - \tau| \mid c] \geq |c_g - \tau|$ *and* $\mathbb{E}[(\tau - \hat{c}_g)_+ \mid c] \geq (\tau - c_g)_+$*, so the naive CovGap and SF are biased upward.*

When every c_g equals τ, the targets are zero and the naive metrics show only test noise.

Proposition 3.2 (Noise floors)**.** *(a)* Oracle. *If $c_g = \tau$ for all g, then* $\mathbb{E}[$*naive* CovGap$] = \mathrm{mean}_{\mathcal{G}_1} D(m_g, \tau)/m_g$*, where* $D(m, p) = \mathbb{E}|\mathrm{Bin}(m, p) - mp| = 2k\binom{m}{k}p^k(1-p)^{m-k+1}$ *with* $k = \lfloor mp \rfloor + 1$ *(de Moivre), and* $D(m, p)/m \sim \sqrt{2p(1-p)/(\pi m)}$*. In the same case* $\mathbb{E}[$*naive* FracBelow$(t)] = \mathrm{mean}_{\mathcal{G}_1} \mathbb{P}(\mathrm{Bin}(m_g, \tau) < t m_g)$*. (b)* Mondrian. *For split Mondrian conformal prediction with n_g calibration points in group g, exchangeability within groups and a.s. distinct scores, $c_g \sim \mathrm{Beta}(k_g, n_g + 1 - k_g)$ with $k_g = \lceil (n_g + 1)\tau \rceil$ when $k_g \leq n_g$ (Vovk, 2012), independently over g given the n_g. Hence*

$$\mathbb{E}[\textit{naive}\ \mathrm{CovGap}] = \mathop{\mathrm{mean}}_{g} \sum_{h=0}^{m_g} \mathrm{BB}_g(h) \left| \frac{h}{m_g} - \tau \right|,$$

with BB_g *the beta-binomial mass function with parameters* $(m_g, k_g, n_g + 1 - k_g)$.

At $\tau = 0.9$ the oracle floor of CovGap is 0.162, 0.118 and 0.024 for $m = 2$, 5 and 100 (Table 4 in Appendix A). The FracBelow floor is not monotone in m: at $m \leq 6$ any single uncovered point puts $\hat{c}_g$ below 0.85, and the floor peaks there (0.41 at $m = 5$). A reported FracBelow of 0.3 on groups with five test points is *less* than what a perfect method would show. On real test sets the floor is a median 62% of the reported CovGap (Section 7).

4. What the data identify

Proposition 4.1 (Moment identification)**.** *Under the RE model in a stratum where every group has $m_g = m$, the law of the data depends on ν_s only through its first m moments $\mu_r = \int x^r \, d\nu_s$, through an invertible linear map. For any polynomial f of degree $\leq m$, $G_s^{-1} \sum_g \sum_r f_r \, (h_g)_r/(m)_r$ is unbiased for $\int f \, d\tilde{\nu}_s$ with no RE model, where $(x)_r$ is the falling factorial.*

Threshold functionals are not polynomials, and they are not identified.

Theorem 4.2 (Threshold functionals are not identified)**.** *Consider a stratum with $m_g \leq m$ under the RE model. (a) Let ν, ν' share their first m moments and let $F(\tilde{\nu}) \to F(\nu)$ a.s. (as for* FracBelow(t) *and for quantiles at continuity points). Then for any estimator T and $\varepsilon > 0$,* $\liminf_{G_s \to \infty} \max_{\nu, \nu'} \mathbb{P}\big(|T - F(\tilde{\nu})| \geq |F(\nu) - F(\nu')|/2 - \varepsilon\big) \geq 1/2$*. (b) For $m \leq 2$ with some groups having $m_g = 2$ (so that the mean μ and the variance $\sigma^2 > 0$ are identified), $t < \mu$, $\sigma^2 < (\mu - t)(1 - \mu)$ and $\mu + \sigma^2/(\mu - t) \leq 1$, the identified set of $\nu'([0, t])$ is exactly $\big[0,\ \sigma^2/(\sigma^2 + (\mu - t)^2)\big]$. For the strict version $\nu'([0, t))$, which FracBelow uses, it is the same interval with the upper end a supremum that is not attained. If every group has $m_g = 1$, only μ is identified and the set is $[0, (1 - \mu)/(1 - t)]$. (c) For general m, the identified set of $\nu'((-\infty, t])$ is the Chebyshev–Markov–Stieltjes interval (Karlin & Studden, 1966), computed on a grid by the moment LP of Section 5. (d) Under the conditions of (b), let $0 \leq x < t$ and $w = \sigma^2/(\sigma^2 + (\mu - x)^2) \geq 1/G_s$ (the conditions of (b) then hold at x too: $\mu + \sigma^2/(\mu - x) < \mu + \sigma^2/(\mu - t) \leq 1$). For every estimator T of* Worst*,* $\max_{\nu', \nu''} \mathbb{P}\big(|T - \mathrm{Worst}| \geq (t - x)/2\big) \geq (1 - e^{-1})/2$*, where ν', ν'' range over laws with the moments (μ, σ^2).*

For example, $\mu = 0.9$ and $\sigma = 0.03$ give the identified set $[0, 0.265]$ for FracBelow(0.85): a stratum of groups with at most two test points cannot distinguish 0% from 26% of groups below 0.85. With $G_s = 1000$, $x = 0$ and $t = 0.85$, every estimator of the worst-group coverage errs by at least 0.42 with probability at least 0.31 under some law consistent with the data.

CovGap is Lipschitz but not a polynomial, and it is partially identified.

Proposition 4.3 (CovGap: identified set and the RMS bound)**.** *(a) If $m = 1$, the identified set of $\int |x - \tau| \, d\nu'$ is $\big[|\mu - \tau|,\ \mu(1 - \tau) + (1 - \mu)\tau\big]$. (b) If $m = 2$, write $d = \mu - \tau$, $R = \sqrt{\sigma^2 + d^2}$, and assume $R \leq \min(\tau, 1 - \tau)$. The identified set is $[R_-, R]$ with $R_- = \max\{|d|,\ 2R^2 + (2\tau - 1)d\}$. (c) For every vector c and every set of groups,* $\mathrm{mean}\,|c_g - \tau| \leq$ RMS *and* $\mathrm{mean}(\tau - c_g)_+ \leq$ RMS *over that set.*

At $\mu = \tau = 0.9$ the sets are $[0, 0.18]$ for $m = 1$ and $[2\sigma^2, \sigma]$ for $m = 2$. No CovGap estimator is consistent in strata with $m \leq 2$, but by (c) CovGap is always below the RMS gap, which is point-identified for $m \geq 2$ (Proposition 4.1 with $f(x) = (x - \tau)^2$).

5. Inference for the RMS gap

For $g \in \mathcal{G}_2$ let $e_g = (\hat{c}_g - \tau)^2 - \hat{c}_g(1 - \hat{c}_g)/(m_g - 1)$ and $\widehat{\mathrm{MSE}} = \mathrm{mean}_{\mathcal{G}_2} e_g$; this is the debiasing used for quadratic

calibration metrics (Ferro & Fricker, 2012; Kumar et al., 2019; Roelofs et al., 2022). Let $G_2 = |\mathcal{G}_2|$, s the sample standard deviation of the e_g, and $\mathrm{se}_f = G_2^{-1}\big(\sum_{\mathcal{G}_2} \mathrm{Var}(e_g \mid \bar{c}, m_g)\big)^{1/2}$ the standard error under a homogeneous coverage $\bar{c} = (\sum h_g + 1)/(\sum m_g + 2)$, where $\mathrm{Var}(e_g \mid c, m)$ is an explicit degree-4 polynomial in c (Appendix A). The interval is

$$\Big[\sqrt{(\widehat{\mathrm{MSE}} - t\,\mathrm{se})_+},\ \sqrt{(\widehat{\mathrm{MSE}} + t\,\mathrm{se})_+}\Big], \qquad (2)$$

with $\mathrm{se} = \max\{s/\sqrt{G_2}, \mathrm{se}_f\}$ and t the $1-\beta/2$ quantile of Student's t_{G_2-1}.

Theorem 5.1 (Validity). *Under Assumption 2.1: (a) $\mathbb{E}[e_g \mid c] = (c_g - \tau)^2$ exactly. (b) If $G_2 \to \infty$ and $G_2^{-1}\sum \mathrm{Var}(e_g \mid c) \geq v > 0$, then $G_2(\widehat{\mathrm{MSE}} - \mathrm{MSE})/(\sum \mathrm{Var}(e_g \mid c))^{1/2} \Rightarrow N(0,1)$. (c) $\mathbb{E}[s^2] = G_2^{-1}\sum \mathrm{Var}(e_g \mid c) + (G_2-1)^{-1}\sum(\theta_g - \bar{\theta})^2$ with $\theta_g = (c_g - \tau)^2$, so (2) has $\liminf \mathbb{P}(\mathrm{RMS} \in CI) \geq 1 - \beta$.*

No assumption is made on how the c_g arise, and the coupling of groups through D_{cal} is irrelevant because the statement is conditional on D_{cal}.

A one-sided finite-sample upper bound. Theorem 5.1 is asymptotic, and its upper end can fail in finite samples. At $\tau = 0.95$ with many groups of a few test points, e_g is right-skewed: a small group rarely has a miss, and a rare miss gives a large e_g. The estimated standard error is then small exactly when $\widehat{\mathrm{MSE}}$ is low. In our synthetic studies the share of replications with RMS above the upper end exceeded $\beta/2$ significantly (one-sided exact binomial test per cell of 500 replications, Holm over cells) in 119 and 204 of 1344 cells on two grids, while the lower end was rarely missed (Section 7). We therefore bound RMS from above in finite samples. Since e_g is bounded below, $Y_g = \theta_g - e_g$ is bounded above, and its sum concentrates; the variance and range depend on the unknown c, so we take their worst case given the MSE.

Proposition 5.2 (Finite-sample upper bounds). *Let $\theta_{\max} = \max(\tau, 1-\tau)^2$, $\ell = \log(1/\delta)$, $v_m(c) = \mathrm{Var}(e_g \mid c, m_g = m)$ and $a_m = \min_{h \leq m} e(h, m)$. Under Assumption 2.1, for every $c \in [0,1]^G$, every (m_g) and every $G_2 \geq 1$, $\mathbb{P}(\mathrm{RMS} \leq U) \geq 1 - \delta$ for $U = U_1$ and for $U = U_3$, where $U^2 = \sup\{t \in [0, \theta_{\max}] : G_2(t - \widehat{\mathrm{MSE}}) \leq s(t)\}$ ($U = 0$ if the set is empty) with:*

- (Bernstein) $s_1(t) = \sqrt{2V^*(t)\ell} + b^*(t)\ell/3$, *where* $V^*(t) = \min_{\lambda \in \Lambda}\{\lambda G_2 t + \sum_{\mathcal{G}_2} \varphi_{m_g}(\lambda)\}$ *with* $\varphi_m(\lambda) = \max_c\{v_m(c) - \lambda(c-\tau)^2\}$, *and* $b^*(t) = \min(G_2 t, \theta_{\max}) + \max_{\mathcal{G}_2}(-a_{m_g})$;
- (Chernoff, exact binomial MGF) $s_3(t) = \min_{(\lambda,\mu)\in\Gamma}\{\ell + \sum_{\mathcal{G}_2} \psi_{m_g}(\lambda,\mu) + \mu G_2 t\}/\lambda$, *where* $\psi_m(\lambda,\mu) = \sup_c\{(\lambda-\mu)(c-\tau)^2 + \log \mathbb{E}_c e^{-\lambda e(h,m)}\}$.

Here $\Lambda \ni 0$ and $\Gamma \subset (0,\infty) \times [0,\infty)$ are finite sets fixed in advance, and the statement holds with φ, ψ replaced by any upper bounds.

Both follow from a concentration inequality for $\sum Y_g$ and weak duality on the constraint $\mathrm{mean}_{\mathcal{G}_2} \theta_g = t$; the feasible sets are intervals, so U is found by bisection (Appendix A). ψ_m is nonincreasing in m because e is an order-2 U-statistic and U-statistics decrease in convex order (Lemma A.1); we compute it as a rigorous upper bound by branch and bound, and reuse it across data sets. Their price is width, from the worst case over c at the boundary value t; whether a bound valid for every c can be much tighter is open (Appendix A.5).

Comparing two methods. Methods are compared on the same test points, so their group counts are dependent. For methods A and B let $d_g = e_g^A - e_g^B$ on $\mathcal{G}_2$, $\Delta = \mathrm{MSE}_A - \mathrm{MSE}_B$, $\hat{\Delta} = \mathrm{mean}_{\mathcal{G}_2} d_g$, and s_d the sample standard deviation of the d_g.

Proposition 5.3 (Paired comparison). *Suppose test points are i.i.d. given both calibrations, so the coverage indicators of A and B on a test point are an i.i.d. pair within each group. Given both calibrations and m: (a) $\mathbb{E}\hat{\Delta} = \Delta$ exactly; (b) the d_g are independent across groups and $\mathbb{E}s_d^2 \geq G_2^{-1}\sum_{\mathcal{G}_2} \mathrm{Var} d_g$; (c) if $G_2 \to \infty$ with $G_2^{-1}\sum \mathrm{Var} d_g \geq v > 0$, the interval $\hat{\Delta} \pm t\, s_d/\sqrt{G_2}$ has* $\liminf$ *coverage $\geq 1 - \beta$.*

The within-group dependence between A and B never needs to be modelled. The comparison is stated for the MSE because differences of CovGap are not identified in small strata (Proposition 4.3). The limit in (c) can be slow: when a method covers almost always in groups with few test points, the d_g are skewed and the interval under-covers (Section 7).

How short can an interval be?

Theorem 5.4 (Length lower bound). *Fix $m_1, \ldots, m_G$ and let C be an interval-valued procedure with $\mathbb{P}_c(\mathrm{RMS}(c) \in C) \geq 1 - \beta$ for every $c \in [0,1]^G$. For $S \subseteq \mathcal{G}_2$ and $\eta \in (0,1)$ let $\rho_S^2 = \tau(1-\tau)\sqrt{2\log(1+4\eta^2)/\sum_{g\in S} m_g^2}$ (with $\tau \pm \rho_S \in [0,1]$). Then at $c \equiv \tau$, $\mathbb{E}_\tau[\mathrm{length}(C)] \geq (1 - 2\beta - \eta)\sqrt{|S|/G_2}\,\rho_S$. For homogeneous m and $S = \mathcal{G}_2$ this is of order $\sqrt{\tau(1-\tau)/m}\, G_2^{-1/4}$.*

The $G^{-1/4}$ rate is the critical separation for testing binomial homogeneity (Chhor & Carpentier, 2022; Kania et al., 2025); Theorem 5.4 states it as a length bound for RMS intervals with heterogeneous m_g. The interval (2) attains the rate up to constants and a $(\log m_{\max})^{1/4}$ factor (Proposition A.2 in Appendix A).

Table 1. Test-set design: the number of groups G_2 at which naive $\mathrm{CovGap}_{\mathcal{G}_2} > U$ in 80% of 400 replications, interpolated in $\log G_2$ between grid points $25 \cdot 2^{k/4}$, when every group has m test points and exact coverage τ. U_0: upper end of the asymptotic interval (2), not valid in finite samples. Curves in the supplement. Run `design_fine/20261013T194157Z-51204d3`.

	$\tau = 0.90$			$\tau = 0.95$		
m	U_0	U_1	U_3	U_0	U_1	U_3
2	120	741	611	387	2,558	2,094
3	44	442	292	136	1,279	803
5	32	366	180	46	804	355
10	49	641	242	28	831	239
20	43	826	192	50	1,802	318
50	37	1,534	162	32	2,719	237
100	38	2,599	163	35	5,199	233

Certifying overstatement, and test-set size. By Proposition 4.3(c), CovGap and SF over $\mathcal{G}_2$ are at most RMS, so an upper bound on RMS also bounds them.

Corollary 5.5 (Certificate; design rule)**.** *(a) With $U \in \{U_1, U_3\}$, $\mathbb{P}(\mathrm{CovGap}_{\mathcal{G}_2} \le U,\ \mathrm{SF}_{\mathcal{G}_2} \le U) \ge 1 - \delta$ for every c and every G_2. Hence "naive $\mathrm{CovGap}_{\mathcal{G}_2} > U$" certifies at level δ that the reported CovGap overstates the realized one. Over $\mathcal{G}_1$, $\mathrm{CovGap}_{\mathcal{G}_1} \le \frac{G_2}{G_1} U + \big(1 - \frac{G_2}{G_1}\big) \max(\tau, 1-\tau)$. (b) For homogeneous m, distinguishing $\mathrm{RMS} = \rho$ from 0 requires, up to constants, $m \gtrsim \tau(1-\tau)/(\rho^2\sqrt{G})$ test points per group, and the asymptotic interval (2) achieves it.*

The certificate is a by-product: by Proposition 3.1 the naive CovGap is biased upward anyway, and what matters more is how large the realized gap can be, which U answers. The rate in (b) is for the asymptotic interval; the finite-sample bounds need more groups (Table 1). At exact validity and $m \ge 5$, U_3 certifies with probability 0.8 from 160–360 groups, while U_1 needs 370–5,200, more as m grows because its variance proxy is the worst case; with $m \le 3$ both need 290–2,600. When coverage varies across groups the naive CovGap overstates less, and certification needs more groups, up to more than 10^5 at $m = 100$ and $\tau = 0.95$; mixed profiles of m_g are harder still (Appendix B.9).

Bounds for the other functionals. For CovGap and $\mathrm{FracBelow}(t)$ we report moment-LP bounds: per stratum of m_g, simultaneous confidence boxes for the compound moments $\mathrm{mean}_s\, c_g^r$ (from Proposition 4.1, $r \le \min(m_{\min,s}, 8)$), then the range of the functional over all laws on a grid with moments in the boxes. They cover the compound functional whenever the boxes do, with no RE model; under the RE model they shrink to the identified sets of Section 4.

Point estimates. For CovGap and SF we use a stratified plug-in: in each stratum of m_g ($[1,4]$, $[5,19]$, $[20,99]$, $[100,\infty)$) we fit the Kiefer–Wolfowitz NPMLE of the mixing law of c_g (Kiefer & Wolfowitz, 1956) on a grid of 301 points by EM, combine the strata by group counts, and integrate the functional. For a stratum with homogeneous m, the NPMLE has $W_1 = O(1/m)$ for $m = O(\log G_s)$ (Vinayak et al., 2019), which is minimax (Tian et al., 2017); since CovGap and SF are 1-Lipschitz, the same rate holds for them. We do not have a guarantee for heterogeneous m_g within a stratum; there the estimator is supported by the experiments only.

6. Auditing individual groups

A practitioner may want the list of groups that are truly under-covered. Write $H_g : c_g \ge t$ for $g \in \mathcal{G}_1$, with $t = \tau - 0.05$, and let $p_g = \mathbb{P}(\mathrm{Bin}(m_g, t) \le h_g)$ be the exact one-sided p-value. Our audit is BH (Benjamini & Hochberg, 1995) at level q on the p_g.

Proposition 6.1 (Finite-sample FDR control)**.** *Under Assumption 2.1, conditionally on D_{cal} and m: the p_g are independent and each null p_g is super-uniform, so BH at level q has $\mathrm{FDR} \le q\pi_0$. The same holds for the discrete BH variant of Döhler et al. (2018) and for a cross-fitted weighted BH with weights from the other fold (Genovese et al., 2006) run at $q/2$ per fold.*

The guarantee holds for every vector c. Popular alternatives lack it with many small groups: flagging by $\hat{c}_g < t$ or per-group Clopper–Pearson intervals has no multiplicity control, a plug-in local fdr depends on the mixing law near t, which is not identified (Theorem 4.2), and the normal approximation of Cherian & Candès (2024) needs group probabilities bounded away from zero.

The test layer also limits power.

Proposition 6.2 (Power ceiling)**.** *BH and its discrete variant flag g only if $p_g \le q$, so $\mathbb{P}(g \text{ flagged}) \le \mathbb{P}_{c_g}\big(F_{\mathrm{Bin}(m_g,t)}(h_g) \le q\big)$, the power of a single level-q exact test, for every G. The smallest attainable p-value is $(1-t)^{m_g}$.*

At $c_g = 0.75$, $t = 0.85$ and $m_g = 10$ the ceiling is 0.224. An audit that flags no group is therefore no evidence that small groups are covered. Discrete and weighted BH were valid in our design stage but added no power over BH (Appendix B).

7. Experiments

Estimators, grids, seeds, comparators and pass criteria were pre-registered before any number was computed, and the real-data pool was held out during development and read once. Criteria and outcomes are in Table 2, runs and seeds in Appendix B.

Table 2. Pre-registered criteria and outcomes, and post hoc analyses. Criteria (Holm over cells at 0.05): point estimation, share of cells where our estimator beats the better comparator (pass $\geq 2/3$); intervals, pooled coverage and cells rejecting coverage ≥ 0.95 (minimum cell); auditing, pooled FDR at $q = 0.1$ and share of cells with power at least that of Cherian & Candès (2024) (C&C). Each study used a fresh grid (1344 cells; for the paired criteria 2016, 3360 and 4704 configuration–pair cells, of which 1168 and 449 had at least 50 replications kept by the rules). Post hoc analyses were designed after the results they use were seen and have no criterion.

Check	Result	Met
Synthetic study		
CovGap estimate	better in 87% of cells	yes
SF estimate	better in 90% of cells	yes
RMS interval (2)	0.975; 17 cells rej. (min 0.874)	no
LP bounds	1.000; 0 cells rej.	yes
Audit: FDR	0.0014; 0 cells rej.	yes
Audit: power	$\geq$ C&C in 73% of cells	yes
$[L, U_1]$, new grid	0.997; 0 cells rej. (min 0.974)	yes
U_1 only, new grid	share above 0.000; 0 rej.	yes
Shared test points		
U_3 only	share above 4×10^{-6}; 0 rej.	yes
Paired interval	0.960; 182 rej. (min 0.762)	no
Rule $f_1 \leq 0.05$ for it	0.971; 3 rej. (min 0.892)	no
Rule and $G_2 \geq 200$	0.980; 0 rej. (min 0.926)	yes
Finite-sample paired	1.000; 0 cells rej.	yes
Permutation nulls	reject 2.4%; 0 rej.	yes
Real data		
Naive/est. ≥ 1.5	3.22, 1.63, 1.59	yes
Certified by U_1	0 of 7 methods	no
Post hoc or exploratory		
Model bootstrap	below asymptotic	–
U_3 on real data	certifies 0 of 51,800	–
Leader changes, paired	separated in $\leq 4.5\%$	–
Exact validity, real m	U_3 certifies in $\leq 17\%$	–
Layer (a), dev pool	descriptive; Appendix B.8	–

Synthetic study: design. Scores are $|\sigma_g \varepsilon|$ with ε standard normal (*scale* family) or Student-t_{ν_g} (*shape* family), $\log \sigma_g$ Gaussian with standard deviation ω, and group frequencies Zipf with tail γ, so the exact c_g given D_{cal} is known. The grid crosses $G \in \{200, 800, 3000\}$, $\gamma \in \{0.7, 1.3\}$, $\omega \in \{0.08, 0.25\}$, $n_{\text{cal}} \in \{8, 40\}$k, $n_{\text{test}} \in \{8, 30, 120\}$k and $\alpha \in \{0.1, 0.05\}$ (288 configurations). 48 semi-synthetic configurations use the group frequencies and sample sizes of six real datasets. Each configuration is evaluated with four group-conditional methods (standard split, Mondrian, and shrinkage of group thresholds towards the pooled one at two strengths), giving 1344 cells of 500 replications each. Comparators are the naive plug-in and a beta-binomial empirical-Bayes plug-in for point estimates, and the simultaneous bootstrap audit of Cherian & Candès (2024) for auditing.

Synthetic study: results. Table 2 gives the outcomes. The median ratio of our absolute error to the better comparator's is 0.41 for CovGap and 0.48 for SF (post hoc, 0.32 against g-modeling (Efron, 2016) and 0.17 against the LP midpoint; Appendix B.12); ours is significantly worse in 7.7% of cells for CovGap, 17.3% among Mondrian cells. The RMS interval missed almost always above its upper end, from the right skew described in Section 5. We replaced that end once, by a second pre-registered procedure whose candidates were fixed in advance and selected on the seen grid (Appendix B.4); on a fresh grid the selected U_1 was never exceeded in 672,000 replications, at a median 1.8 ($\alpha = 0.1$) and 2.4 ($\alpha = 0.05$) times the true RMS. The LP bounds are valid but wide for FracBelow (median width 0.36–0.77). BH on exact binomial p-values has FDR 0.0014 and power 0.045 (0.017 for Cherian & Candès (2024); 0.009 in many-small cells, the ceiling of Proposition 6.2); popular alternatives reach FDR 0.19–0.62 (Appendix B.3).

Real data: design. Seven datasets with pre-specified groups give 13 dataset–score cells (Table 3): five regression datasets with CQR and absolute-residual scores, and class-conditional classification on DBpedia and CIFAR-100 with APS and LAC scores. A quarter of each dataset was held out during all development and read once. Each of 200 random splits draws a calibration set of 1.5k–10k points and uses the rest as test set. The seven methods are standard split, Mondrian, shrinkage at three strengths, clustered conformal (Ding et al., 2023) and conditional conformal with group indicators (Gibbs et al., 2025), on LightGBM, TF-IDF logistic regression and ViT-S probes at $\alpha \in \{0.1, 0.05\}$, plus a second base model at $\alpha = 0.1$. From the development data we pre-registered two predictions for ACS, airlines and house sales (CQR, $\alpha = 0.1$): in each cell, the median over the non-Mondrian methods of naive over estimated CovGap is at least 1.5; and at least 4 of the 7 methods are certified, with mean naive RMS above the upper bound.

Real data: results. The first prediction held, with CovGap ratios 3.22, 1.63 and 1.59. The second did not: no method has mean naive RMS above U_1 in any of the three cells. We had based it on the asymptotic upper end of (2), which is below the naive RMS for 6 of 7 methods in each cell but is not valid in finite samples; U_1 replaced it after the synthetic study, before the real data were read. The naive CovGap never exceeds U_1 in the 51,800 (run, method, split) evaluations; U_1 is 1.1–3.7 times the naive RMS.

Since the debiased MSE is unbiased and the naive one exceeds it by the test noise in expectation (Proposition 3.1 and Theorem 5.1), one minus their ratio estimates the test-noise share without a model for the c_g (Table 3, "noise"). It is a median 41% in the primary $\alpha = 0.1$ runs and ranges from 5% to essentially all. In ACS it is 1.01: for every method except Mondrian the mean debiased RMS is at most 0.015 against a naive RMS of 0.082–0.084 (Table 19). The oracle floor agrees for CovGap: a method with exact coverage in every group would show a median 62% of the

Table 3. Real data, primary runs at $\alpha = 0.1$ (200 splits each). $|\mathcal{G}_1|$: groups with a test point; small: share of them with $m_g < 20$. Medians over the 7 methods of: noise, the test-noise share of the naive MSE, $1 - \widehat{\text{MSE}}/\overline{\text{naive MSE}}$ (model-free; above 1 when the debiased MSE is slightly negative); naive CovGap over our CovGap estimate; the oracle floor (Proposition 3.2a, from the realized m) over the naive CovGap; U_1 over the naive RMS. r_{K}: Kendall correlation of the method orders by naive and by estimated CovGap. Runs `final_d/20261013T022808Z-3726b4b` and `summarize_p4r/20261013T135857Z-906c785`; all 37 runs in Appendix B.10.

Dataset (groups)	Score	$\lvert\mathcal{G}_1\rvert$	small	noise	naive / est.	floor / naive	U_1 / naive RMS	r_{K}
ACS (state×race)	CQR	408	0.35	1.01	3.20	0.96	1.60	0.90
Airlines (origin)	CQR	352	0.27	0.61	1.61	0.65	1.38	1.00
Airlines (origin)	abs	352	0.27	0.88	1.87	0.76	1.65	0.62
Zurich (stop)	CQR	1528	0.06	0.41	1.25	0.54	1.17	1.00
Zurich (stop)	abs	1528	0.06	0.22	1.15	0.44	1.16	1.00
DBpedia (L3 class)	APS	219	0.00	0.29	1.27	0.57	2.45	1.00
DBpedia (L3 class)	LAC	219	0.00	0.05	1.03	0.20	1.23	1.00
CIFAR-100 (class)	APS	100	0.00	0.37	1.32	0.64	2.31	0.90
CIFAR-100 (class)	LAC	100	0.00	0.30	1.21	0.57	2.17	0.71
Diamonds (color×clarity)	CQR	56	0.10	0.44	1.34	0.62	2.89	0.90
Diamonds (color×clarity)	abs	56	0.10	0.26	1.11	0.42	2.05	0.90
House sales (zipcode)	CQR	70	0.08	0.57	1.59	0.77	2.82	0.90
House sales (zipcode)	abs	70	0.09	0.55	1.27	0.67	2.67	1.00

reported CovGap (96% in ACS). The naive CovGap is 1.03–3.20 times our estimate, which is a point in an identified set when some $m_g \leq 2$ (Proposition 4.3).

Rankings change less: the median r_{K} is 0.90, and the method with the smallest CovGap differs between the naive and the estimated metric in 6 of 26 primary runs and in none of the 11 robustness runs. Mondrian is the exception to inflation: its naive and estimated CovGap agree (ACS: 0.084 and 0.084) because its realized coverages do vary (Proposition 3.2b). Against standard split conformal in ACS, the naive metric makes it look 1.6 times worse, our estimate 5.3 times. Marginal coverage is at least nominal minus 0.005 in 253 of 259 (run, method) pairs; the exceptions, interval lengths and set sizes are in Appendix B.10.

Shared test points: U_3 and paired comparisons. Three further pre-registered synthetic studies drew one test set per replication for all methods, as in a real evaluation (Appendices B.5 to B.7). U_3 was exceeded in 3 of 672,000 replications, against $\delta = 0.025$, at 1.25–2.07 times the true RMS against 1.69–3.02 for U_1. The paired interval undercovered in 182 of 2016 cells (minimum 0.76), 175 of them where the median group has at most 20 test points. $\hat{\Delta}$ is unbiased there. The misses come from pairs in which one method covers almost always in groups with few test points (Mondrian at $\alpha = 0.05$): a rare miss moves $\hat{\Delta}$ and s_d together, and in the samples without such a miss the d_g look unskewed. A rule chosen on these draws, to use the interval only if $f_1 \leq 0.05$, the share of groups in $\mathcal{G}_2$ with $m_g \leq 20$ and $\hat{c}_g = 1$ for either method, failed on a new grid in 3 of 1168 cells, all with 150 groups at $\alpha = 0.05$ (coverage 0.89–0.91). Adding $G_2 \geq 200$, chosen on the draws seen so far, held on a third grid: coverage 0.980, no cell rejected, but it keeps 9% of replications and none in the many-small regime. The exact nulls held their level: pooled over all their replications the test rejected in 5.6% against a reflected partner $2\tau - c_g$, and in 2.4% (no cell rejected) against a permutation partner that covers almost always in the same small groups, so the failure needs one method near 1 and the other not. A finite-sample paired interval never missed but was a median 50 times wider (Appendix A.5), and a model bootstrap covered less often than the asymptotic interval (Appendix B.7).

Real data, post hoc. These analyses were designed after the real-data results were seen (Appendices B.6, B.7 and B.11). On the logged counts, U_3 is a median 0.38–0.97 times U_1 per run and still above the naive CovGap and RMS in all 51,800 evaluations. Simulated on these test counts, even an exactly valid method is certified by U_3 in at most 17% of test sets (Appendix B.9). The refined rule keeps method pairs only in Zurich and DBpedia, where the paired test separates 1–15 of the kept pairs per run, and none of the 6 leader changes, which the test separates in at most 4.5% of splits. Mondrian's larger gap is mostly over-coverage: its estimated shortfall is below standard split's in 24 of 26 primary runs (ratio 0.03–0.92), while its estimated CovGap is up to 5.5 times larger (Table 15).

Layer (a) and one-sided gaps. On the development pool (ACS and airlines; $K = 5$ disjoint blocks with the study's n_{cal}; descriptive, Appendix B.8), Mondrian has the largest gap at layer (a) too (MSE 5.7 and 5.1 against 0.4 and 1.4×10^{-3} for standard split; only differences between methods are estimated precisely), and 98–99% of it is over-coverage. Its cross-fitted layer-(a) shortfall is about zero, as its guarantee implies, against 3.0 and 16.1×10^{-3} for

standard split, so by one-sided gap the ranking reverses. The other methods match standard split in MSE at layer (a); the excess of shrinkage at layer (b) is calibration variance.

8. Related work

Conditional coverage and its evaluation. Group- and class-conditional conformal methods (Vovk, 2012; Ding et al., 2023; Gibbs et al., 2025; Ding et al., 2026; Bhattacharyya et al., 2026) are compared through per-group empirical coverage on a test set. Coverage as a function of x is estimated by classifying the coverage indicator (Braun et al., 2026; Zhou et al., 2026) or audited over data-driven slabs (Cauchois et al., 2021); Min et al. (2026) show that the coverage–length metric can be gamed. Ding et al. (2023) note that classwise coverage given a calibration draw is Beta-distributed (our layer b). We found no conformal paper that corrects group metrics for the binomial test layer, derives their noise floor under exact validity, or gives identification limits for them.

Statistical tools. Our RMS estimator is the debiasing used for the Brier-score reliability term (Bröcker, 2012; Ferro & Fricker, 2012) and squared calibration error (Kumar et al., 2019; Roelofs et al., 2022). Shrinkage estimates of subgroup performance (Ji et al., 2020; Miller et al., 2021) are under-dispersed and bias tail functionals toward zero. Estimating the distribution of success probabilities from a few trials per unit is classical (Teicher, 1963; Kiefer & Wolfowitz, 1956; Koenker & Mizera, 2014), with optimal W_1 rates by Tian et al. (2017); Vinayak et al. (2019) and heterogeneous trial profiles in Lee & Kennedy (2026). Moment identification and bounds on the share of units below a cut were used to bound discrimination by Kline & Walters (2021); Kline et al. (2022); Gu & Koenker (2023) treat ranking and selection of many noisy units as compound decisions, the decision problem behind flagging or ranking groups. Many small-domain proportions are the subject of small-area estimation (Rao & Molina, 2015), and our test-noise share is one minus a reliability ratio of classical test theory (Lord & Novick, 1968). Cherian & Candès (2024) give simultaneous bootstrap bounds over rich collections of groups; our audit controls the FDR with exact p-values, which remain valid with one test point per group.

9. Discussion and limitations

A group-coverage gap cannot be interpreted without the test counts behind it. In our runs the ranking of methods usually survived the correction but the size of the gaps did not, and threshold metrics are not identified for small groups.

Recommendations. (1) Report the oracle noise floor (Proposition 3.2) next to any naive group metric; it needs only the test counts m_g. (2) Report the debiased RMS gap, or the estimated CovGap with the caveat of Proposition 4.3, rather than the naive CovGap; do not report the fraction below a cut or the worst group for groups with few test points, or report the LP bounds. (3) Compare methods with the paired interval of Proposition 5.3 only when $G_2 \geq 200$ and at most 5% of the groups in $\mathcal{G}_2$ have $m_g \leq 20$ and $\hat{c}_g = 1$ for either method, a rule confirmed on a fresh grid that counts groups, not test points per group (Appendix B.7). With many small groups no comparison we tried is both valid and informative: the asymptotic interval over-rejects, invisibly in the sample, and the finite-sample one cannot separate methods. (4) Report one-sided gaps such as the shortfall next to symmetric ones: over-coverage costs length, under-coverage costs a group its guarantee. (5) Bound how large the realized gap can be with U_3, and check before evaluating whether the test counts could certify at all (Table 1, Appendix B.9). (6) Flag individual groups with BH on exact binomial p-values, and expect few flags for small groups (Proposition 6.2).

Limitations. The targets are conditional on one calibration draw (layer b); our layer-(a) estimates come from the development pool only. Our CovGap and SF estimates have guarantees only within strata of equal m_g, and they were worse than the better comparator in 7.7% of synthetic cells. The asymptotic RMS and paired intervals under-cover with small groups and high coverage. The finite-sample bounds are valid for every c and pay for it with width: on our real test sets they certified no overstatement. Range-based adaptive bounds are wider still (Proposition A.4); at exact validity the median of any valid bound is at least a third to a half of U_3's on homogeneous profiles with $G_2 = 400$, and whether one can certify at these sizes is open (Appendix A.5). All guarantees rest on Assumption 2.1, which holds only approximately in our data, and none covers distribution shift.

Impact Statement

Group-conditional coverage is used to argue that a predictor treats demographic or other groups fairly. The usual metrics are symmetric: they count over-coverage, which costs longer intervals, like under-coverage, which leaves a group below its guarantee. Our results show that these metrics can overstate how unevenly a method covers groups, and that unevenness is not the same as unfairness: Mondrian conformal prediction has the largest gap in our data, but mostly from over-coverage, and its estimated shortfall below the target is smaller than standard split's in most runs. One-sided measures are the relevant ones for fairness claims, and they are only partially identified when groups have few test points. The tools here reduce false alarms about individual groups; their low power for small groups means that the absence of a flag is not evidence of adequate coverage. All

datasets are public; the ACS microdata are de-identified by the U.S. Census Bureau.

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A. Proofs

A.1. Proofs for Section 3

Proof of Proposition 3.1. (a) $\mathbb{E}[\hat{c}_g \mid c] = c_g$ and $\mathrm{Var}(\hat{c}_g \mid c) = c_g(1-c_g)/m_g$; expand the square. (b) Jensen's inequality for the convex maps $x \mapsto |x-\tau|$ and $x \mapsto (\tau - x)_+$. □

Proof of Proposition 3.2. (a) Substitute $c_g = \tau$. De Moivre's formula for the mean absolute deviation of the binomial is classical (Diaconis & Zabell, 1991); the asymptotic form is the central limit theorem for $\mathbb{E}|Z|$. (b) The Beta law is cited. Given the n_g, the threshold of group g depends only on that group's calibration points, which are disjoint across groups, so the c_g are independent. Since $h_g \mid c_g \sim \mathrm{Bin}(m_g, c_g)$, marginally h_g is beta-binomial. □

Table 4. Oracle noise floor (Proposition 3.2a) at $\tau = 0.9$: expected naive metrics of a method with exact group coverage 0.9 in every group. Run `check_theory_D/20261012T155044Z-ae0b210`.

m	$\mathbb{E}$[naive CovGap]	$\mathbb{E}$[naive FracBelow(0.85)]
1	0.180	0.10
2	0.162	0.19
3	0.146	0.27
5	0.118	0.41
10	0.070	0.26
20	0.051	0.13
100	0.024	0.04

A.2. Proofs for Section 4

Proof of Proposition 4.1. The Bernstein polynomials $\{\binom{m}{j}x^j(1-x)^{m-j}\}_{j=0}^m$ form a basis of the polynomials of degree $\le m$, so the map from $(1, \mu_1, \dots, \mu_m)$ to $(\mathbb{P}(h=j))_{j=0}^m$ is a change of basis. Unbiasedness follows from $\mathbb{E}[(h)_r \mid c] = (m)_r c^r$. □

Proof of Theorem 4.2. (a) By Proposition 4.1, ν and ν' induce the same data law. With $\delta = |F(\nu) - F(\nu')|$, the events $\{|T - F(\nu)| < \delta/2\}$ and $\{|T - F(\nu')| < \delta/2\}$ are disjoint, so one has probability at most $1/2$ under the common law. Replacing $F(\nu)$ by $F(\tilde{\nu})$ costs ε for large G_s by the a.s. convergence.

(b) *Upper end.* Cantelli's inequality gives $\mathbb{P}(X \le t) \le \sigma^2/(\sigma^2 + (\mu - t)^2)$ for $t < \mu$. It is attained by the two-point law with mass $w = \sigma^2/(\sigma^2 + (\mu-t)^2)$ at t and $1-w$ at $b = \mu + \sigma^2/(\mu - t)$: its mean is $wt + (1-w)b = \mu$ and its variance is $w(1-w)(b-t)^2 = \sigma^2$, and $b \le 1$ by assumption. *Lower end.* The two-point law on $\{t', 1\}$ with mass $(1-\mu)/(1-t')$ at t' has mean μ and variance $(\mu - t')(1-\mu)$, which tends to $(\mu - t)(1-\mu) > \sigma^2$ as $t' \downarrow t$; fix $t' \in (t, \mu)$ with $(\mu - t')(1-\mu) \ge \sigma^2$. The mixture $\lambda \cdot (\text{two-point}) + (1-\lambda)\delta_\mu$ has mean μ and variance $\lambda(\mu - t')(1-\mu)$; choose λ to match σ^2. It puts no mass on $[0, t]$. *Interval.* The constraint set is convex and the functional is linear in ν'.

(c) Karlin & Studden (1966), Ch. III–IV; see also Krein & Nudelman (1977), Ch. III.

(d) Let ν' be the two-point law of the upper end of (b) with x in place of t (mass w at x; its other point $\mu + \sigma^2/(\mu - x)$ is below $\mu + \sigma^2/(\mu - t) \le 1$ because $x < t$, and $x \ge 0$), and ν'' the law of the lower end of (b) at t, which puts no mass on $[0, t]$. Both have moments (μ, σ^2), so the data law, and hence the law of T, is the same under both. Under ν'', Worst $> t$ a.s., so accuracy within $(t-x)/2$ requires $T > (x+t)/2$. Under ν', Worst $= x$ with probability $p = 1 - (1-w)^{G_s} \ge 1 - e^{-1}$, and on that event accuracy requires $T < (x+t)/2$. The two accuracy probabilities therefore sum to at most $1 + (1-p)$, so one of them is at most $1 - p/2$. □

Proof of Proposition 4.3. (a) The lower end is Jensen's inequality, attained by δ_μ. Since $x \mapsto |x - \tau|$ is convex, for a fixed mean its expectation is maximized by the law on $\{0, 1\}$ with mass μ at 1, which gives $\mu(1-\tau) + (1-\mu)\tau$.

(b) *Upper end.* Cauchy–Schwarz gives $\int |x - \tau| \le R$, with equality iff $|x - \tau|$ is constant a.s.; the law on $\tau \pm R$ with $\mathbb{P}(\tau + R) = (1 + d/R)/2$ matches both moments. *Lower end.* Let $Y = X - \tau \in [-A, B]$ with $A = \tau$, $B = 1 - \tau$, $u = \mathbb{E}Y_+$ and $v = \mathbb{E}Y_- = u - d$. Since $Y_+^2 \le BY_+$ and $Y_-^2 \le AY_-$, $R^2 = \mathbb{E}Y^2 \le Bu + Av = u - Ad$, hence

$\mathbb{E}|Y| = 2u - d \geq 2R^2 + (2A-1)d$; Jensen gives $\mathbb{E}|Y| \geq |d|$. *Attainment.* If $R^2 \leq Bd$ ($d \geq 0$) or $R^2 \leq A|d|$ ($d \leq 0$), a law on one side of τ with these moments exists (Bhatia & Davis, 2000) and $\mathbb{E}|Y| = |d|$. Otherwise take masses $b = (R^2 + Ad)/B$ at B, $a = (R^2 - Bd)/A$ at $-A$ and $1-a-b$ at 0. In this case $a, b \geq 0$. Using $A + B = 1$, $a + b = \big(R^2 + (A-B)d\big)/(AB) \leq \big(R^2 + |A-B|R\big)/(AB) \leq \big(\min(A,B)^2 + |A-B|\min(A,B)\big)/(AB) = 1$, by $|d| \leq R \leq \min(A,B)$. The law has mean $bB - aA = d$ and second moment $bB^2 + aA^2 = R^2$, and $\mathbb{E}|Y| = bB + aA = 2R^2 + (A-B)d$. *Interval.* As in Theorem 4.2(b).

(c) Cauchy–Schwarz, and $(\tau - x)_+ \leq |x - \tau|$. □

A.3. Proofs for Section 5

Proof of Theorem 5.1. (a) $\mathbb{E}[\hat{c}^2] = c^2 + c(1-c)/m$ and $\mathbb{E}[\hat{c}(1-\hat{c})] = c(1-c)(1-1/m)$, so $\mathbb{E}[\hat{c}(1-\hat{c})]/(m-1) = c(1-c)/m$; combine with Proposition 3.1(a). Writing $Z = \hat{c} - c$, $e = \text{const} + bZ + \gamma Z^2$ with $b = 2(c-\tau) - (1-2c)/(m-1)$ and $\gamma = m/(m-1)$, so $\mathrm{Var}(e \mid c) = b^2\mu_2 + \gamma^2(\mu_4 - \mu_2^2) + 2b\gamma\mu_3$ with the binomial central moments $\mu_2 = c(1-c)/m$, $\mu_3 = c(1-c)(1-2c)/m^2$, $\mu_4 = c(1-c)(1 + 3(m-2)c(1-c))/m^3$. (b) The summands are independent given c and bounded ($|e_g| \leq 1.25$), and their variances sum to infinity, so Lindeberg's condition holds. (c) For independent X_g with means θ_g and variances v_g, $\mathbb{E}\sum(X_g - \bar{X})^2 = (1 - 1/G)\sum v_g + \sum(\theta_g - \bar{\theta})^2$. Bounded summands give $s^2 - \mathbb{E}s^2 \to 0$ in probability, so $s^2/(G_2^{-1}\sum v_g) \geq 1 - o_P(1)$. The maximum with se_f only widens the interval, and clipping at 0 followed by the monotone map $\sqrt{\cdot}$ can only enlarge the set. □

Proof of Proposition 5.2. U_1. Let $Y_g = \theta_g - e_g$, $g \in \mathcal{G}_2$. Given c the Y_g are independent (Assumption 2.1), with $\mathbb{E}Y_g = 0$ (Theorem 5.1(a)) and $\mathrm{Var}Y_g = v_{m_g}(c_g)$. Moreover $Y_g \leq \theta_g - a_{m_g} \leq b(c) := \max_g(\theta_g - a_{m_g})$, and $b(c) \geq 0$ because $a_m \leq \mathbb{E}e_g = \theta_g$. Since $Y_g \leq b$ gives $\mathbb{E}[(Y_g)_+^q] \leq b^{q-2}\mathbb{E}Y_g^2 \leq (q!/2)(b/3)^{q-2}\mathbb{E}Y_g^2$ for $q \geq 3$, Bernstein's inequality in sub-gamma form (Boucheron et al., 2013, Thm. 2.10) gives $\mathbb{P}\big(\sum Y_g \geq \sqrt{2V(c)\ell} + b(c)\ell/3\big) \leq \delta$ with $V(c) = \sum v_{m_g}(c_g)$. Write $t_0 = \mathrm{MSE}(c)$. For every $\lambda \geq 0$, $V(c) = \sum[v_{m_g}(c_g) - \lambda\theta_g] + \lambda G_2 t_0 \leq \sum \varphi_{m_g}(\lambda) + \lambda G_2 t_0$, so $V(c) \leq V^*(t_0)$. Each $\theta_g \geq 0$ and $\sum\theta_g = G_2 t_0$, so $\theta_g \leq \min(G_2 t_0, \theta_{\max})$ and $b(c) \leq b^*(t_0)$. Outside the δ-event, $G_2(t_0 - \widehat{\mathrm{MSE}}) = \sum Y_g < \sqrt{2V^*(t_0)\ell} + b^*(t_0)\ell/3$, so t_0 is feasible and $\mathrm{RMS} = \sqrt{t_0} \leq U_1$. V^* is a minimum of affine functions of t, hence concave, and so are $\sqrt{2V^*\ell}$ and b^*; the feasible set is therefore an interval.

U_3. For $\lambda > 0$, Markov's inequality gives $\mathbb{P}(\sum Y_g \geq s) \leq \exp(-\lambda s + \sum K_{m_g}(\lambda, c_g))$ with $K_m(\lambda, c) = \log \mathbb{E}_c e^{\lambda(\theta - e)} = \lambda(c-\tau)^2 + \log \mathbb{E}_c e^{-\lambda e(h,m)}$. For $\mu \geq 0$, $\sum K = \sum[K - \mu\theta_g] + \mu G_2 t_0 \leq \sum \psi_{m_g}(\lambda, \mu) + \mu G_2 t_0$. Let $s_k(t)$ be the threshold of $(\lambda, \mu)_k \in \Gamma$ computed with the exact ψ, and k^* the minimizer of $s_k(t_0)$; k^* depends only on t_0 and m, so $\mathbb{P}(\sum Y_g \geq s_{k^*}(t_0)) \leq \delta$. With upper bounds $\hat{\psi} \geq \psi$, the computed threshold satisfies $s_3(t_0) = \min_k \hat{s}_k(t_0) \geq s_{k^*}(t_0)$, so $\{G_2(t_0 - \widehat{\mathrm{MSE}}) > s_3(t_0)\} \subseteq \{\sum Y_g \geq s_{k^*}(t_0)\}$. This holds even if the effort spent tightening $\hat{\psi}$ depends on the data. s_3 is a minimum of affine functions of t, hence concave. □

Lemma A.1 (Convex order in m)**.** *For fixed c, let $U_m = e(h, m)$ with $h \sim \mathrm{Bin}(m, c)$. Then $\mathbb{E}\phi(U_{m'}) \leq \mathbb{E}\phi(U_m)$ for every convex ϕ and $m' \geq m \geq 2$. In particular $\psi_{m'} \leq \psi_m$ and $v_{m'} \leq v_m$.*

Proof. With x_i i.i.d. Bernoulli(c), $e = (m(m-1))^{-1}\sum_{i \neq j}(x_i - \tau)(x_j - \tau)$ is a U-statistic of order 2. $U_{m'}$ is the average of U_m over the m-subsets of the m' observations, which are exchangeable given their symmetric σ-field $\mathcal{G}$, so $U_{m'} = \mathbb{E}[U_m(x_1, \ldots, x_m) \mid \mathcal{G}]$; apply Jensen's inequality. □

Computing $\hat{\psi}$. We need only $\hat{\psi} \geq \psi$. With $w_h = \exp(-\lambda e_h - \max_h(-\lambda e_h))$, which is unimodal in h because e is convex in h, we bound $\log \mathbb{E}_c w$ on each interval $[c_0, c_1]$ by the smallest of three upper bounds and maximize the quadratic term exactly: (i) with $w = w^{\uparrow} + w^{\downarrow} - w(h^*)$ ($w^{\uparrow}$ nondecreasing, $w^{\downarrow}$ nonincreasing) and stochastic monotonicity of $\mathrm{Bin}(m, c)$ in c, $\mathbb{E}_c w \leq \mathbb{E}_{c_1} w^{\uparrow} + \mathbb{E}_{c_0} w^{\downarrow} - w(h^*)$; (ii) $p(c) = \mathbb{E}_c w$ is a polynomial with $-p'' \leq m(m-1)\max(-\Delta^2 w)_+$, so $p \leq \max(p(c_0), p(c_1)) + (c_1 - c_0)^2 m(m-1)\max(-\Delta^2 w)_+/8$; (iii) $\frac{d}{dc}\log \mathbb{E}_c w \in [-m/(1-c), m/c]$. Binomial sums use exact tail masses and floating-point margins. Branch and bound refines intervals; for $m > 200$ we use $\hat{\psi}$ at a geometric grid point $\leq m$ (Lemma A.1). Tests check $\hat{\psi}$ against a dense grid and the coverage of U_3 by exact enumeration for $G_2 \leq 3$.

Proof of Proposition 5.3. (a) is Theorem 5.1(a) for each method: h_g^A and h_g^B are each binomial whatever their joint law. (b) Groups use disjoint test points, which are independent given the calibrations; the expectation of s_d^2 decomposes as in Theorem 5.1(c). (c) As Theorem 5.1(b, c), with $|d_g| \leq 5/4$ because $e \in [-1/4, 1]$. □

Layer (a) from disjoint blocks (Section 2). Let blocks $k = 1, \dots, K$ use disjoint calibration and test sets drawn from the population, so that the calibrations, and hence the coverage vectors $c^{(k)}$, are i.i.d. across blocks and independent of the test counts. For a group with a test point in every block, $\mathbb{E}\hat{c}_g^{(k)} = \mathbb{E}[\mathbb{E}(\hat{c}_g^{(k)} \mid c^{(k)})] = \mathbb{E}c_g$ by Assumption 2.1, and for $k \neq l$ independence gives $\mathbb{E}[(\hat{c}_g^{(k)} - \tau)(\hat{c}_g^{(l)} - \tau)] = (\mathbb{E}c_g - \tau)^2$. Averaged over groups, the cross-block mean is a U-statistic of order 2 in the blocks with kernel $\mathrm{mean}_g(\hat{c}_g^{(k)} - \tau)(\hat{c}_g^{(l)} - \tau)$. Its jackknife variance over blocks is consistent as $K \to \infty$ (Arvesen, 1969) and approximate for small K. Groups within a block may be dependent through the shared calibration; this does not matter, because the blocks are the resampled units.

Proposition A.2 (Rate optimality). *At $c \equiv \tau$, the MSE-scale half-width of (2) and the MSE-scale lower bound of Theorem 5.4 are within a factor $\sqrt{4K\kappa(1 + 1/(2\tau(1-\tau)))}/\sqrt{2\log(1+4\eta^2)}$, where $K \leq 1 + \log_2 m_{\max}$ and $\kappa = \kappa(\tau)$ is a numerical constant ($\kappa(0.9) = \kappa(0.95) = 2.00$). On the RMS scale the factor is the fourth root of this. Numerically, at $c \equiv \tau$ the length of (2) is 3.0–3.8 times the bound of Theorem 5.4 for homogeneous $m \in \{2, \dots, 20\}$ and $G \in \{100, \dots, 6400\}$, and 3.6–4.1 times for Zipf profiles (run `check_theory_D/20261012T155044Z-ae0b210`).*

Proof of Theorem 5.4. Let ε_g, $g \in S$, be i.i.d. Rademacher, and set $c_g = \tau + \rho_S \varepsilon_g$ on S and τ elsewhere. Every such c has $\mathrm{RMS} = r := \sqrt{|S|/G_2}\,\rho_S$. Let P_{mix} be the ε-mixture of the data laws. For a Bernoulli with base p and perturbations δ_1, δ_2, $\sum_x p_{\delta_1}(x)p_{\delta_2}(x)/p(x) = 1 + \delta_1\delta_2/(p(1-p))$. Tensorizing over trials and groups,

$$1 + \chi^2(P_{\mathrm{mix}} \| P_\tau) = \prod_{g \in S} \mathbb{E}\big[(1 + a\varepsilon_g\varepsilon_g')^{m_g}\big] = \prod_{g \in S} \frac{(1+a)^{m_g} + (1-a)^{m_g}}{2},$$

with $a = \rho_S^2/(\tau(1-\tau))$ and ε' an independent copy. Each factor is at most $\cosh(m_g a) \leq \exp(m_g^2 a^2/2)$, so $\chi^2 \leq \exp(a^2 \sum_S m_g^2/2) - 1 = 4\eta^2$ and $\mathrm{TV} \leq \frac{1}{2}\sqrt{\chi^2} \leq \eta$. Coverage under each alternative gives $P_{\mathrm{mix}}(r \in C) \geq 1 - \beta$, hence $P_\tau(r \in C) \geq 1 - \beta - \eta$; coverage at τ gives $P_\tau(0 \in C) \geq 1 - \beta$. On the intersection, of probability at least $1 - 2\beta - \eta$, $C \supseteq [0, r]$. □

Proof of Proposition A.2. Write $pq = \tau(1-\tau)$. (i) At $c_g = \tau$, $\mathrm{Var}(e_g \mid \tau) \leq \kappa\,(pq/m_g^3 + (pq)^2/m_g^2)$ with $\kappa(\tau)$ computed numerically from the exact variance. (ii) Since $m_g \geq 2$, $pq/m^3 \leq (pq)^2/m^2 \cdot 1/(2pq)$, so $\mathrm{se}_f \leq pq\,\big(\kappa(1 + 1/(2pq)) \sum_{\mathcal{G}_2} m_g^{-2}\big)^{1/2}/G_2$. (iii) With dyadic layers $L_k = \{g : 2^{k-1} < m_g \leq 2^k\}$ of sizes n_k and K non-empty layers, $\sum_{\mathcal{G}_2} m_g^{-2} \leq \sum_k n_k 4^{1-k} \leq 4K \max_k n_k/4^k \leq 4K \max_S |S|^2/\sum_S m_g^2$. (iv) The MSE-scale lower bound of Theorem 5.4 is $r^2 = pq\, c_\eta \max_S(|S|/G_2)/(\sum_S m_g^2)^{1/2}$ with $c_\eta = \sqrt{2\log(1+4\eta^2)}$; comparing with (ii)–(iii) gives the factor. The sample-sd branch is of the same order by Theorem 5.1(c). □

Proof of Corollary 5.5. (a) By Proposition 4.3(c), $\mathrm{CovGap}_{\mathcal{G}_2} \leq \mathrm{RMS}$ and $\mathrm{SF}_{\mathcal{G}_2} \leq \mathrm{RMS}$, and $\mathrm{RMS} \leq U$ fails with probability at most δ (Proposition 5.2). For $\mathcal{G}_1$, split the mean into $\mathcal{G}_2$ and the groups with $m_g = 1$, on which $|c_g - \tau| \leq \max(\tau, 1-\tau)$. (b) Necessity is the proof of Theorem 5.4; sufficiency is Theorem 5.1 with step (ii) of the proof of Proposition A.2. □

A.4. Proofs for Section 6

Proof of Proposition 6.1. Given D_{cal} and m, the h_g are independent (Assumption 2.1), hence so are the p_g. $\mathbb{P}_c(\mathrm{Bin}(m, c) \leq h)$ is decreasing in c, and at $c = t$ the p-value is uniform on its support, so null p-values are super-uniform. BH then has $\mathrm{FDR} \leq q\pi_0$ (Benjamini & Hochberg, 1995). The discrete variant is Döhler et al. (2018), Thm. 3.1. For the cross-fitted version, each fold's weights are independent of its p-values and average to one, so each fold's weighted BH at $q/2$ has FDR at most $q/2$ (Genovese et al., 2006); since $V/\max(R, 1) \leq V_A/\max(R_A, 1) + V_B/\max(R_B, 1)$, the union has FDR at most q. □

Proof of Proposition 6.2. BH's critical values are at most q, and the discrete variant's are too when the null CDFs are uniform on their supports, so a flagged group has $p_g \leq q$. The smallest attainable p-value is $\mathbb{P}(\mathrm{Bin}(m_g, t) \leq 0) = (1-t)^{m_g}$. □

A.5. How far can a finite-sample upper bound go?

Corollary A.3 (One-sided lower bound). *In the setting of Theorem 5.4, let U satisfy $\mathbb{P}_c(\mathrm{RMS}(c) \leq U) \geq 1 - \delta$ for every $c \in [0,1]^G$. Then $\mathbb{P}_\tau(U \geq r_S) \geq 1 - \delta - \eta$ with $r_S = \sqrt{|S|/G_2}\,\rho_S$, and therefore $\mathbb{P}_\tau$(naive $\mathrm{CovGap}_{\mathcal{G}_2} > U$) $\leq$*

$\mathbb{P}_\tau(\textit{naive}\ \mathrm{CovGap}_{\mathcal{G}_2} > r_S) + \delta + \eta$.

Proof. Every alternative in the proof of Theorem 5.4 has $\mathrm{RMS} = r_S$, so $\mathbb{P}_\varepsilon(U \geq r_S) \geq 1 - \delta$ for each sign vector; averaging, $P_{\mathrm{mix}}(U \geq r_S) \geq 1 - \delta$, and $\mathrm{TV}(P_{\mathrm{mix}}, P_\tau) \leq \eta$. Replacing ρ_S by a smaller value keeps $\chi^2 \leq 4\eta^2$. □

We evaluated the bound with S the k groups of $\mathcal{G}_2$ with the smallest m_g, ρ_S capped at $1 - \tau$, r_S maximised over k and the bound minimised over η, on homogeneous profiles ($m \in \{2, \ldots, 100\}$, $G_2 \in \{100, \ldots, 6400\}$) and on the 26 primary real count profiles (run `lb_profiles/20261013T194155Z-51204d3`). There r_S is only 5–27% of the naive CovGap, so the bound exceeds 1 everywhere: this construction cannot rule out certification by some valid bound.

Asymmetric alternatives. The same argument holds for any prior π under which the c_g are independent with a law ν_{m_g} that depends only on m_g: $\mathbb{P}_\tau(U \geq r) \geq 1 - \delta - \kappa(r) - \mathrm{TV}$, with $\kappa(r) = \mathbb{P}_\pi(\mathrm{RMS}(c) < r)$ and $\mathrm{TV} \leq (1 - \mathrm{BC}^2)^{1/2}$, where BC is the product over groups of the Bhattacharyya coefficients between the mixed and the $\mathrm{Bin}(m_g, \tau)$ laws. Then $\mathbb{P}_\tau(\text{naive } \mathrm{CovGap}_{\mathcal{G}_2} > U) \leq \delta + \kappa(r) + \mathrm{TV} + \mathbb{P}_\tau(\text{naive } \mathrm{CovGap}_{\mathcal{G}_2} > r)$, and κ and the last term are bounded by Chernoff bounds with the exact moment generating functions. We optimized over asymmetric two-point laws for each m (step 0.005) and their four-point mixtures, allocating the Hellinger budget across values of m by a common multiplier (`confpre.eval.lower_opt`; run `lb_opt/20261014T212530Z-37b5470`). Asymmetry helps only when the per-group budget is large: at budgets of at most 10^{-3} in $-\log \mathrm{BC}_m$, the regime of hundreds of groups, the best asymmetric law reaches at most 1.22 times the RMS of the symmetric one (less than 1 where our grid of two-point laws is coarser than the symmetric perturbation); at 10^{-2} it reaches up to 5.3 times, by putting a little mass on a coverage far below τ. The bound on the certification probability still exceeds 1 on every homogeneous and real profile. What the construction does give is a floor on the median: at exact validity every valid U has median at least r_{med}, the largest r with $\delta + \kappa(r) + \mathrm{TV} \leq 1/2$. Table 5 compares it with U_3 at $G_2 = 400$: U_3 is within a factor of 2.1–3.0 of what any valid bound must report. On the real profiles r_{med} is 0.50–0.55 (ACS), 0.48–0.60 (airlines) and 0.24–0.33 (Zurich) of the naive CovGap, and at most 0.22 elsewhere (Corollary A.3 gave 5–27%). Whether a valid bound can certify at these sizes, and whether one much tighter than U_3 exists for mixed profiles of m_g, is open.

Table 5. Homogeneous profiles with $G_2 = 400$ groups of m test points, exact validity: the floor r_{med} on the median of every valid upper bound, the median of U_3, and the floor of Proposition A.4 on range-only bounds (probability $1/2$), all relative to the median naive CovGap over $\mathcal{G}_2$ (200 simulated test sets). Run `summarize_r4/20261014T232714Z-a110f10`.

	$\tau = 0.90$			$\tau = 0.95$		
m	r_{med}	U_3	range-only	r_{med}	U_3	range-only
2	0.43	1.02	0.48	0.58	1.62	0.92
3	0.33	0.81	0.53	0.39	1.18	0.97
5	0.31	0.69	0.66	0.33	0.88	1.07
10	0.36	0.77	1.12	0.31	0.76	1.37
20	0.33	0.72	1.51	0.34	0.84	2.29
50	0.30	0.67	2.35	0.28	0.76	3.30
100	0.29	0.69	3.28	0.29	0.79	4.81

Range-based adaptive bounds. For $y \geq -1$ and $\lambda \in [0, 1)$, $\exp(\lambda y - \psi_E(\lambda) y^2) \leq 1 + \lambda y$ with $\psi_E(\lambda) = -\log(1-\lambda) - \lambda$ (Fan et al., 2015, Lemma 4.1). With predictable centering $\hat{e}_g$ (the running mean of the earlier e in a fixed order of the groups), $B = \theta_{\max} + 1/4 \geq e_g - \hat{e}_g$ and $y = (\hat{e}_g - e_g)/B$, the products $\prod_g \exp\left(\lambda(\theta_g - e_g)/B - \psi_E(\lambda)(e_g - \hat{e}_g)^2/B^2\right)$ form a nonnegative supermartingale for every c, so $\mathbb{P}\big(\sum(\theta_g - e_g) \geq [B\ell + \psi_E(\lambda) V/B]/\lambda\big) \leq e^{-\ell}$ with $V = \sum(e_g - \hat{e}_g)^2$. A union bound over 16 fixed values of λ gives an empirical-Bernstein bound U_{EB}, and the same argument applied to d_g gives a finite-sample paired interval, valid for every pair of coverage vectors and every joint law of the two methods on a test point. Both use the realized spread of the e_g instead of the worst case over c, but they pay for the range: e_g can reach $\theta_{\max}$ in any group, so the slack is at least $B\ell/(\lambda_{\max} G_2) \approx 7/G_2$ on the MSE scale. Proposition A.4 shows that every procedure whose validity rests only on independence and the range of the e_g, betting bounds (Waudby-Smith & Ramdas, 2024) included, faces a limit of this kind. U_{EB} visits the groups in index order, fixed before the data; sorting them by m_g in either direction changed it by less than 0.2% on every real profile. In a design probe at exact validity (run `probe_eb/20261013T192232Z-479d9c2`), U_{EB} was 2.6–14 times the naive CovGap on the real count profiles, against 1.1–3.6

for U_3, and never certified. The paired version never missed in the study of Appendix B.6 but was a median 50 times wider than the asymptotic interval.

Proposition A.4 (Range-only bounds)**.** *Let $\mathcal{P}$ be the laws of independent $e_1, \dots, e_{G_2}$ with $e_g \in [a, \theta_{\max}]$, and $\bar{\theta}$ the mean of their means. If $\mathbb{P}(\bar{\theta} \le U^2) \ge 1 - \delta$ for every law in $\mathcal{P}$, then for every $P_0 \in \mathcal{P}$ and $\varepsilon \in (\delta, 1)$, $P_0\big(U^2 < \bar{\theta}_0 + (\theta_{\max} - \bar{\theta}_0)\, p\big) \le \varepsilon$ with $p = 1 - (\delta/\varepsilon)^{1/G_2}$.*

Proof. Let P_1 make each e_g independently equal to $\theta_{\max}$ with probability p and otherwise draw it from P_0. Then $P_1 \in \mathcal{P}$ with $\bar{\theta}_1 = \bar{\theta}_0 + p(\theta_{\max} - \bar{\theta}_0)$. For $A = \{U^2 < \bar{\theta}_1\}$, $\delta \ge P_1(A) \ge (1-p)^{G_2} P_0(A) = (\delta/\varepsilon) P_0(A)$, since with probability $(1-p)^{G_2}$ no e_g is replaced and the data are then distributed as under P_0. □

At exact validity ($\bar{\theta}_0 = 0$) any such bound therefore exceeds $\big(\theta_{\max}(1 - (\delta/\varepsilon)^{1/G_2})\big)^{1/2} \approx (\theta_{\max} \log(\varepsilon/\delta)/G_2)^{1/2}$ with probability at least $1 - \varepsilon$, whatever the test counts. U_1 and U_3 escape this because they use the binomial law of e_g, under which $e_g = \theta_{\max}$ requires $h_g = 0$. With $\varepsilon = 1/2$ the floor is 1.5–11.7 times the median naive CovGap on the real profiles, so no range-only bound certifies there in more than about half of the test sets; on homogeneous profiles with $m \ge 5$ it stays above the naive CovGap up to hundreds or thousands of groups, where U_3 already certifies (Table 5).

B. Experimental details

B.1. Pre-registration and disclosures

- Estimators, grids, seeds, comparators, pass criteria and the real-data predictions were frozen before any confirmatory number was computed. Synthetic replication r of configuration k uses the random stream $(66000 + k, r)$; real split r of run k uses $(53000 + k, r)$. Smoke tests use seeds offset by 900,000.
- One timing-only smoke test printed a few per-configuration estimates at the seeds then planned for the synthetic study. The seed base was moved before the freeze, regardless of the values, so none of those draws are part of the confirmatory data.
- A bug in one auditing candidate (the weight filter of the cross-fitted weighted BH) was found and fixed before the freeze. The design stage below was rerun in full.
- Confirmatory analyses (Table 2): `analyze_p4/20261012T222216Z-5490cfb` (synthetic study), `analyze_p4s2/20261013T013856Z-b340cac` (repair), `analyze_r1/20261013T172049Z-9206add` (shared test points), `analyze_r2/20261013T200831Z-509f35f` (paired rule and exact nulls), `analyze_r3/20261014T225505Z-46dcad4` (refined paired rule and permutation nulls) and `analyze_p4/20261013T094117Z-62767b9` (real data).
- Two procedures failed their criteria and were followed up under new pre-registrations on fresh grids, with the failed results kept in Table 2: the RMS upper bound, repaired once (Appendix B.4), and the paired interval (Appendix B.5), for which a rule of use then failed (Appendix B.6) and a refined rule passed (Appendix B.7). The second real-data prediction failed and was not replaced. Apart from these follow-ups, no estimator, criterion or prediction was changed after one of its confirmatory numbers was seen.
- The plans of the studies in Table 2, and the code of each of their confirmatory runs, reached a public code host before the runs started; the server-side push times are in `provenance.md`. The plans of Appendices B.6 and B.7 were also submitted for OpenTimestamps stamps at the freeze, but the Bitcoin blocks that anchor the stamps came 38 and 4 minutes after those runs ended. The stamps therefore bound how late the plans can have been written; they do not by themselves show that the plans preceded the results.

B.2. Day clustering of coverage (Assumption 2.1)

On the development pool of the airline and Zurich data (20 random calibration draws of 10,000 points, CQR score, LightGBM, $\alpha = 0.1$; the rest of the pool is the test set), we computed for each group g and calendar day d the counts (m_{gd}, h_{gd}) and the within-group dispersion statistic $\phi = \sum_g \sum_d (h_{gd} - m_{gd}\hat{c}_g)^2 / (m_{gd}\hat{c}_g(1-\hat{c}_g)) \big/ \sum_g (D_g - 1)$, and the same statistic after permuting the day labels within each group (ϕ_{perm}, which absorbs small-count effects). Medians over draws, run `diag_dispersion/20261013T142434Z-9f2b89e`:

Data	Method	ϕ	ϕ_{perm}	ϕ/ϕ_{perm}	test points per group-day
Airlines	standard	1.029	1.004	1.024	5.3
Airlines	Mondrian	1.025	1.003	1.022	5.3
Zurich	standard	1.030	1.009	1.020	6.3
Zurich	Mondrian	1.037	1.004	1.037	6.3

Coverage clusters by day only slightly. Our test sets are the rest of the pool and contain its group-days whole, so this is also the variance inflation in our own evaluation, about 2–4%: Assumption 2.1 holds approximately, and the floor and the debiasing are understated by about that fraction, which makes the noise-share estimates slightly low.

B.3. Auditing: comparators and design-stage choice

On the confirmatory synthetic grid (pooled over cells; run `analyze_p4/20261012T222216Z-5490cfb`), the FDR at $q = 0.1$ is 0.62 for flagging by $\hat{c}_g < t$, 0.19 for a plug-in local fdr, 0.20 for per-group Clopper–Pearson intervals and 0.24 for BH on the asymptotic p-values of Cherian & Candès (2024), against 0.0014 for BH on exact binomial p-values.

The auditing procedure was chosen before the confirmatory study, on previously seen synthetic configurations (50 replications), by a rule fixed in advance:

- among the candidates with pooled FDR at most $q + 2\,\text{SE}$, take the one with the highest median TPR over cells with at least one non-null group;
- candidates within 0.005 count as tied, and the simplest wins (BH, then discrete BH, then weighted BH).

Discrete BH had median TPR 0.0163, BH was tied with it, and the weighted BH had 0.0091, so BH was chosen. Runs: `design_audit/20261012T162246Z-3cefce6` and `analyze_design_audit/20261012T163703Z-b951f3e`.

B.4. Repair of the RMS upper bound

Three candidates were fixed before any was computed:

- U_0, the upper end of (2), kept as reference;
- U_1 of Proposition 5.2;
- U_2, a bootstrap-t over groups (999 resamples) studentized as in (2).

They were compared on the already-seen synthetic grid at fresh seeds, with 200 replications per cell. A candidate qualified if it met both of these criteria in all 1344 cells:

- the one-sided criterion, share above $U \le 0.025$;
- the two-sided criterion of Table 2 for $[L, U]$.

U_1 was to be selected if it qualified and its excess over the debiased RMS was at most 1.5 times the smallest qualifying excess. Otherwise the narrowest qualifying candidate would be selected. Table 6 gives the selection stage; U_1 was the only candidate that qualified. The confirmatory test on a fresh grid is in Table 2.

Table 6. Selection stage of the repair (seen grid, fresh seeds; 1344 cells × 200 replications). Run `analyze_p4s2/20261013T001324Z-0b04cd0`.

	share above U	cells rejected (one-sided)	coverage of $[L, U]$ (min cell)	cells rejected (two-sided)	excess
U_0	0.0226	38	0.975 (0.865)	4	0.012
U_1	0.0000	0	0.997 (0.965)	0	0.069
U_2	0.0619	405	0.935 (0.630)	202	0.012

B.5. Shared test points: U_3 and the paired interval

This third synthetic study was pre-registered (plan, gates and readings) before any of its numbers were computed. Seeds are $(69000 + k, r)$, and the four methods are evaluated on test points drawn once per replication, so their counts are dependent as in a real evaluation. The grid crosses $G \in \{250, 1000, 4000\}$, $\gamma \in \{0.8, 1.2\}$, $\omega \in \{0.12, 0.30\}$, $n_{\text{cal}} \in \{10, 50\}$k, $n_{\text{test}} \in \{10, 40, 160\}$k and both α (288 configurations), plus 48 semi-synthetic configurations. One attempt crashed on a cache-file race before producing output and was rerun in full. Analysis run `analyze_r1/20261013T172049Z-9206add`.

Why the paired interval under-covers (descriptive, after the gate; run `diag_r1_paired/20261013T172241Z-c4f4415`). The standardized bias of $\hat{\Delta}$ across the 2016 cells has mean 0.03 and s.d. 1.05, with 3 cells beyond $|z| > 3$, about what chance gives. In the worst cells the s.d. of $\hat{\Delta} - \Delta$ is 1.00–1.04 times the RMS standard error. The misses there are all on one side, with skewness of $\hat{\Delta} - \Delta$ near ± 0.8 and correlation between $\hat{\Delta} - \Delta$ and s_d near ± 0.9. These cells are Mondrian pairs at $\alpha = 0.05$ with at most 10 calibration points per group on average, $n_{\text{test}} = 10$k, $\gamma = 1.2$ and $\omega = 0.12$; their coverage is 0.76–0.79.

Can the sample flag it? (descriptive; run `diag_r1_skew/20261013T183844Z-e45b02a`, which regenerates the draws from their seeds and reproduces every logged estimate; bins fixed before the run). Two statistics computable from one test set do not flag the failure: the sample skewness of the mean, $\hat{\gamma} = \text{skew}(d_g)/\sqrt{G_2}$, and the effective number of groups $(\sum(d_g - \bar{d})^2)^2 / \sum(d_g - \bar{d})^4$. Coverage is lowest where $|\hat{\gamma}|$ is smallest (Table 7) and where the effective number of groups is largest (0.933 above 100, against 0.968 below 10). The failing replications are those in which the rare miss did not occur, and such samples look unskewed. A warning therefore has to come from the design, few test points per group and a method whose coverage is near 1, not from the realized d_g.

Table 7. Shared test points, descriptive: coverage of the paired 95% interval by $|\hat{\gamma}|$, computed in each replication from its own d_g, by regime; last column: share of all replications. Run `summarize_r2/20261013T184553Z-4cd2cb1`.

$\lvert\hat{\gamma}\rvert$	other	many small	all	share
< 0.05	0.975	0.882	0.933	0.08
0.05–0.1	0.975	0.899	0.937	0.09
0.1–0.2	0.979	0.922	0.947	0.22
0.2–0.4	0.990	0.945	0.965	0.38
≥ 0.4	0.997	0.977	0.984	0.23

Table 8. Shared test points, by regime: medians over cells of U_1 and U_3 over the true RMS and of U_3/U_1; means over cells of the share of replications with naive $\text{CovGap}_{\mathcal{G}_2} > U$. "Many small": the median group has at most 20 expected test points. U_0 is not valid in finite samples. The true RMS never exceeded U_1.

		width			certification rate		
α	regime	U_1/RMS	U_3/RMS	U_3/U_1	U_0	U_1	U_3
0.10	pure, many small	1.85	1.72	0.94	0.71	0.26	0.33
0.10	pure, other	1.69	1.25	0.78	0.14	0.00	0.00
0.10	semi-synthetic	2.10	1.64	0.81	0.06	0.00	0.00
0.05	pure, many small	2.42	2.07	0.89	0.57	0.05	0.11
0.05	pure, other	2.23	1.42	0.63	0.14	0.00	0.00
0.05	semi-synthetic	3.02	2.08	0.72	0.06	0.00	0.00

B.6. A rule for the paired interval, and exact nulls

Pre-registered before any of its numbers were computed; the plan was also stamped with OpenTimestamps at the freeze (`provenance.md`). Grid: $G \in \{150, 600, 2500\}$, $\gamma \in \{0.9, 1.1\}$, $\omega \in \{0.10, 0.25\}$, $n_{\text{cal}} \in \{20, 80\}$k, $n_{\text{test}} \in \{5, 25, 125\}$k, both α and both families (288 configurations), plus 48 semi-synthetic ones; 500 replications with seeds $(85000 + k, r)$; test points shared by all methods. Each method A is paired with its reflection A', $c_g^{A'} = 2\tau - c_g^A$ when that lies in $[0, 1]$ and c_g^A otherwise, realized by its exact threshold on the same test points, so $\Delta = 0$ exactly. The rule ($f_1 \leq 0.05$) was chosen among five candidates on the draws of Appendix B.5 (run `explore_r1_rule/20261013T192556Z-a874c55`); there it kept 26% of replications with no cell rejected. Runs `r2_synth/20261013T200416Z-4e757d4` and `analyze_r2/20261013T200831Z-509f35f`.

Table 9. Shared test points: coverage of the paired 95% interval for $\mathrm{MSE}_A - \mathrm{MSE}_B$ (top) and share of replications with $p < 0.05$ for $\Delta = 0$ by the size of $|\Delta|$ (bottom), by α and regime.

	$\alpha = 0.10$		$\alpha = 0.05$	
Pair	other	many small	other	many small
Mondrian–shrink $\kappa = 10$	0.978	0.943	0.967	0.908
Mondrian–shrink $\kappa = 200$	0.986	0.946	0.975	0.913
shrink $\kappa = 10$–shrink $\kappa = 200$	0.994	0.963	0.992	0.958
standard–Mondrian	0.988	0.948	0.979	0.914
standard–shrink $\kappa = 10$	0.995	0.965	0.995	0.960
standard–shrink $\kappa = 200$	0.992	0.942	0.988	0.932
$\lvert\Delta\rvert$	share with $p < 0.05$			
$< 10^{-4}$	0.22	0.36	0.32	0.30
10^{-4}–10^{-3}	0.61	0.40	0.64	0.39
10^{-3}–$3 \cdot 10^{-3}$	0.84	0.50	0.90	0.56
$3 \cdot 10^{-3}$–10^{-2}	0.97	0.79	1.00	0.61

Gates. The rule: pooled miss rate 0.029 on the kept replications (threshold 0.051), but 3 of the 1168 cells with at least 50 kept replications reject coverage ≥ 0.95. All three are pure configurations with $G = 150$, $n_{\text{test}} = 25$k and $\alpha = 0.05$: Mondrian against shrinkage ($\kappa = 10$) twice and the exact null of standard split conformal, with coverage 0.89–0.91. The rule is therefore not confirmed. The finite-sample paired interval: no miss in 1,680,000 replications. Table 10 gives the rejection rates under the exact nulls.

Table 10. Exact nulls ($\Delta = 0$): share of replications with $p < 0.05$ for the asymptotic paired test, on all replications, on those the rule keeps and on those it drops; share kept; and the rejection rate of the finite-sample interval. "Many small": the median group has at most 20 expected test points.

α	regime	all	kept	dropped	kept share	finite-sample
0.10	other	0.052	0.050	0.060	0.83	0.000
0.10	many small	0.050	–	0.050	0.00	0.000
0.05	other	0.056	0.051	0.076	0.80	0.000
0.05	many small	0.067	–	0.067	0.00	0.000

Real data, post hoc (run `r2_real/20261013T200931Z-ad1c44c`; logged counts only). Table 11 gives, per primary run at $\alpha = 0.1$, the pairs separated in at least half of the splits by the asymptotic test (Holm over the 21 pairs), without the Mondrian pairs, the pairs the rule keeps in at least half of the splits and those of them separated, and the pairs the finite-sample interval separates. Of the six leader changes, the rule keeps only the CIFAR-100 one, and the finite-sample interval separates none.

Table 11. Post hoc, primary runs at $\alpha = 0.1$: method pairs (of 21) separated by the asymptotic paired test in at least half of the splits (all; without Mondrian), pairs the rule keeps and those of them separated, and pairs separated by the finite-sample interval.

Dataset (groups)	Score	all	no Mondrian	kept	kept, sep.	finite-sample
ACS (state×race)	CQR	13	7	0	0	0
Airlines (origin)	CQR	9	9	0	0	0
Airlines (origin)	abs	10	4	0	0	0
Zurich (stop)	CQR	14	8	15	8	0
Zurich (stop)	abs	17	11	15	11	0
DBpedia (L3 class)	APS	12	6	21	12	0
DBpedia (L3 class)	LAC	15	10	21	15	0
CIFAR-100 (class)	APS	2	0	21	2	0
CIFAR-100 (class)	LAC	1	0	21	1	0
Diamonds (color×clarity)	CQR	3	3	15	3	0
Diamonds (color×clarity)	abs	0	0	15	0	0
House sales (zipcode)	CQR	0	0	15	0	0
House sales (zipcode)	abs	1	0	15	0	0

B.7. A refined rule, permutation nulls and a model bootstrap

Before the plan (exploratory, on the already-seen draws of Appendices B.5 and B.6).

- *Model bootstrap.* We tried a parametric bootstrap-t (`confpre.eval.paired_boot`). Per method, a stratified NPMLE of the law of c_g gives each group a posterior; in each of 499 bootstrap worlds the coverages are drawn from the posteriors (or fixed at the posterior means), the counts are redrawn on the observed m_g with nested test points, and the quantiles of the studentized error calibrate the interval. On the 73 R1-S configurations with a failing cell, 24 others and the three failing R2-S configurations (100 replications each; runs `probe_boot/20261014T204445Z-7b1f583` and `...210819Z-7a993e6`), it covered less often than the asymptotic interval almost everywhere (Table 13).

- *Why* (run `diag_boot/20261014T212116Z-19bd28f`). The model is fitted to the same counts, so it misses the rare misses exactly when the sample does. Where the interval fails (Mondrian pairs, many small groups, $\alpha = 0.05$) the true studentized error has quantiles $(-2.53, 3.36)$, and the bootstrap gives medians of about $(-1.5, 2.6)$ to $(-1.5, 2.9)$. Elsewhere the bootstrap quantiles move against the realized error (correlation -0.2 to -0.56). In a controlled case with 400 groups of 10 test points, the asymptotic interval covered 0.953 and the bootstrap 0.887–0.933. We did not take it to a confirmatory study.

- *Rule search* (run `explore_r3_rule/20261014T211840Z-c1eda8a`; an attempt at `...211817Z` crashed before output). Adding $G_2 \geq g$ to the rule $f_1 \leq 0.05$, $g = 200$ was the smallest value in $\{0, 150, 200, 250, 300, 400, 500\}$ with no rejected cell on the draws of Appendices B.5 and B.6 together. The failing cells of Appendix B.6 had 150 groups and about 167 test points per group on average, many of the groups small, so the count of groups, not the average count per group, is what the rule needs.

Confirmatory study. Pre-registered (`docs/followup/plan_R3.md`, stamped with OpenTimestamps at the freeze) before any of its numbers were computed. Grid: $G \in \{120, 500, 2000\}$, $\gamma \in \{0.85, 1.15\}$, $\omega \in \{0.09, 0.20\}$, $n_{\text{cal}} \in \{15, 60\}$k, $n_{\text{test}} \in \{6, 20, 100\}$k, both α and both families (288 configurations), plus 48 semi-synthetic ones; 500 replications with seeds $(93000 + k, r)$; test points shared by all methods. Besides the reflection, each method A has a permutation partner: the groups with $m_g \geq 2$ are sorted by calibration and test count and adjacent groups swap their coverages, so $\Delta = 0$ exactly and, when A covers almost always in small groups, so does its partner, in the same groups. Runs `r3_synth/20261014T225051Z-5233d07` and `analyze_r3/20261014T225505Z-46dcad4`.

Gates. Both passed. The refined rule ($f_1 \leq 0.05$ and $G_2 \geq 200$): pooled miss rate 0.020 on the kept replications (threshold 0.051), no rejection among 449 cells with at least 50 kept replications, largest cell miss rate 0.074. It kept 9.2% of all replications, none in the many-small regime ($f_1 \leq 0.05$ alone kept 30.7%). The permutation nulls: rejection rate 0.024 pooled, no rejection among 1344 cells, largest 0.068. Table 12 breaks the exact nulls down.

Table 12. Exact nulls on the grid of Appendix B.7: share of replications with $p < 0.05$ for the asymptotic paired test, on all replications, on those the refined rule keeps and on those it drops, and the share kept. Run `summarize_r4/20261014T232714Z-a110f10`.

null	α	regime	all	kept	dropped	kept share
permutation	0.10	other	0.015	0.010	0.016	0.20
permutation	0.10	many small	0.036	–	0.036	0.00
permutation	0.05	other	0.016	0.009	0.017	0.19
permutation	0.05	many small	0.030	–	0.030	0.00
reflection	0.10	other	0.051	0.049	0.052	0.21
reflection	0.10	many small	0.054	–	0.054	0.00
reflection	0.05	other	0.055	0.051	0.056	0.19
reflection	0.05	many small	0.072	–	0.072	0.00

Real data, post hoc (run `r3_real/20261014T225525Z-46dcad4`; logged outputs only). The refined rule keeps method pairs only in Zurich (the 15 pairs without Mondrian; 1–11 of them separated per run) and DBpedia (all 21; 5–15 separated). It keeps none in ACS, airlines, CIFAR-100 (100 groups), diamonds and house sales, and none of the six leader changes.

Table 13. Exploratory, on already-seen draws: coverage of the 95% paired interval, asymptotic and model bootstrap with the coverages redrawn from the posteriors or fixed at the posterior means, by the configurations' status in the earlier studies and by kind of pair. For the nulls, coverage is one minus the rejection rate. Run `summarize_r4/20261014T232714Z-a110f10`.

configurations	pairs	asymptotic	bootstrap (draws)	bootstrap (means)
with a failing cell	Mondrian pairs	0.894	0.908	0.910
with a failing cell	other pairs	0.931	0.893	0.899
with a failing cell	permutation nulls	0.967	0.828	0.834
with a failing cell	reflection nulls	0.938	0.897	0.915
other	Mondrian pairs	0.967	0.906	0.906
other	other pairs	0.985	0.946	0.951
other	permutation nulls	0.979	0.871	0.884
other	reflection nulls	0.948	0.876	0.905

B.8. Layer (a) and one-sided gaps

Descriptive, on the development pool only (run `layer_a/20261014T225524Z-46dcad4`; it reproduces every column of the earlier run `...20261013T202116Z-300f894` exactly and adds the one-sided ones). ACS (state×race) and airlines (origin), CQR, LightGBM, $\alpha = 0.1$. Each of 40 random partitions splits the pool into $K = 5$ disjoint blocks; each block calibrates the seven methods on $n_{\text{cal}} = 10{,}000$ points, the size used in the real-data study, and uses the rest of the block (73k and 40k points) as test set. Over the groups with at least two test points in every block (267 and 249), Table 14 gives the mean over blocks of the debiased MSE (layer b, which estimates $\text{mean}_g[(\mathbb{E}c_g - \tau)^2 + \text{Var}c_g]$), the cross-block estimator of $\text{mean}_g(\mathbb{E}c_g - \tau)^2$ (layer a) with its leave-one-block-out jackknife standard error, and the difference from standard split conformal, averaged over the partitions. The standard errors of the layer-(a) levels are as large as the levels themselves for most methods, so only the differences between methods, whose standard errors are small, are informative; for standard split the layer-(a) estimate exceeds the layer-(b) one, which cannot hold in expectation, by less than one standard error.

One-sided gaps. Symmetric metrics count over-coverage like under-coverage. Two cross-fitted statistics separate them at layer (a): for each block, a group is classed as under-covered when its mean empirical coverage over the *other* blocks is below τ, and SF_a averages $\tau - \hat{c}_g$ over the classed groups (it estimates $\text{mean}_g(\tau - \mathbb{E}c_g)\mathbb{P}(\text{classed})$, close to $\text{mean}_g(\tau - \mathbb{E}c_g)_+$; it can be slightly negative when no group is under-covered); the under-coverage share is the part of the cross-block estimator from pairs of blocks whose complement classes the group as under-covered. Mondrian's layer-(a) shortfall is about zero (-1.7 and -1.8, standard errors 0.8 and 1.0, $\times 10^{-3}$), as its guarantee $\mathbb{E}c_g \geq \tau$ implies, against 3.0 and 16.1 (standard errors 2.2 and 3.3) for standard split; the differences are -4.6 and -18.0 (standard errors 2.3 and 3.6). Only 1–2% of Mondrian's layer-(a) MSE comes from under-coverage, against 82–85% for standard split. The ranking by one-sided gap thus reverses the ranking by MSE for Mondrian. At layer (b), the NPMLE shortfall gives the same order (Table 14), and on the final pool Mondrian's estimated shortfall is below standard split's in 24 of 26 primary runs while its estimated CovGap is up to 5.5 times larger (Table 15; run `r3_real/20261014T225525Z-46dcad4`, logged outputs only). The shortfall is only partially identified when groups have at most two test points (Proposition 4.3); the NPMLE gives a point in that set.

B.9. Design rule: heterogeneous coverage and real test counts

As Table 1 (coarse grid, run `design_rule/20261013T172142Z-9206add`), with $c_g = \tau + 0.03 z_g$ (z_g standard normal, clipped to $[0, 1]$), so the true RMS is about 0.03. The naive CovGap is then partly real gap, and certification needs more groups; at $m = 100$ and $\tau = 0.95$ no grid value reaches 80%.

Real test-count profiles (post hoc; only the counts m are read). The function `cert_rates` in `confpre/eval/design.py` of the supplement estimates the certification rate for any profile of test counts. Applied to the real test sets (Table 16), it shows that an exactly valid method is certified by U_3 in at most 17% of test sets (CIFAR-100, 100 classes with similar counts) and in at most 2% elsewhere; U_1 never certifies, and with heterogeneous coverage neither does U_3. The asymptotic U_0, which is not valid, would "certify" it in 8–100%. Mixed profiles are much harder than Table 1 suggests: Zurich has about 1525 groups, most with many test points, and is never certified. On these test sets the certificate is out of reach whether or not the methods have real gaps.

Table 14. Layer (a) and one-sided gaps on the development pool ($\times 10^{-3}$, means over 40 partitions; jackknife standard errors in parentheses): MSE at layers (b) and (a), difference from standard split at layer (a), the cross-fitted layer-(a) shortfall SF_a, the layer-(b) shortfall (stratified NPMLE, mean over blocks) and the share of the layer-(a) MSE from under-covered groups. Run `summarize_r4/20261014T232714Z-a110f10`.

		MSE			shortfall		
Data	Method	layer (b)	layer (a)	(a) − standard	layer (a)	layer (b)	under share
ACS	standard	0.30	0.41 (0.39)	–	2.96	8.20	0.85
ACS	Mondrian	7.10	5.74 (0.11)	+5.32 (0.41)	-1.65	5.41	0.01
ACS	shrink $\kappa = 10$	0.87	0.47 (0.39)	+0.06 (0.05)	0.97	10.48	0.74
ACS	shrink $\kappa = 50$	0.50	0.43 (0.39)	+0.02 (0.02)	1.37	8.57	0.79
ACS	shrink $\kappa = 200$	0.33	0.41 (0.39)	0.00 (0.01)	1.93	7.83	0.84
ACS	clustered	0.34	0.40 (0.39)	−0.01 (0.06)	2.74	8.70	0.87
ACS	conditional (ridge)	0.30	0.41 (0.39)	−0.01 (0.02)	2.94	8.24	0.86
Airlines	standard	1.19	1.40 (0.55)	–	16.14	20.26	0.82
Airlines	Mondrian	6.59	5.07 (0.15)	+3.66 (0.60)	-1.81	6.36	0.02
Airlines	shrink $\kappa = 10$	1.89	1.40 (0.55)	0.00 (0.08)	11.71	22.28	0.78
Airlines	shrink $\kappa = 50$	1.54	1.39 (0.55)	−0.02 (0.06)	12.17	20.31	0.79
Airlines	shrink $\kappa = 200$	1.32	1.37 (0.55)	−0.04 (0.04)	13.04	19.42	0.80
Airlines	clustered	1.28	1.37 (0.56)	−0.04 (0.14)	15.92	20.64	0.82
Airlines	conditional (ridge)	1.22	1.44 (0.55)	+0.03 (0.09)	16.70	20.97	0.82

Table 15. Final pool, primary runs at $\alpha = 0.1$, post hoc: Mondrian over standard split conformal, medians over the 200 splits, for the naive and estimated CovGap and the naive and estimated shortfall. Run `summarize_r4/20261014T232714Z-a110f10`.

		CovGap		shortfall	
Dataset (groups)	Score	naive	estimated	naive	estimated
ACS (state×race)	CQR	1.57	5.47	0.13	0.42
Airlines (origin)	CQR	1.13	1.87	0.10	0.12
Airlines (origin)	abs	1.30	2.42	0.13	0.22
Zurich (stop)	CQR	1.83	2.30	0.24	0.29
Zurich (stop)	abs	1.45	1.66	0.19	0.21
DBpedia (L3 class)	APS	2.35	3.04	1.40	1.77
DBpedia (L3 class)	LAC	0.68	0.68	0.19	0.19
CIFAR-100 (class)	APS	1.38	1.68	0.81	0.92
CIFAR-100 (class)	LAC	1.18	1.31	0.71	0.74
Diamonds (color×clarity)	CQR	1.34	1.72	0.37	0.41
Diamonds (color×clarity)	abs	0.89	0.94	0.23	0.23
House sales (zipcode)	CQR	1.35	1.87	0.63	0.82
House sales (zipcode)	abs	1.10	1.19	0.50	0.51

B.10. Real data: all runs

Table 18 extends Table 3 to all 37 runs. Table 19 gives per-method coverage, efficiency and estimates in the three cells of the pre-registered predictions. Table 20 summarizes marginal coverage by method. Runs `final_d/20261013T022808Z-3726b4b` and `summarize_p4r/20261013T135857Z-906c785`. Ridge runs of CQR cells use the normalized score $|y - \mu|/\sigma$, because the ridge model has no quantile predictions. Marginal coverage is at least nominal minus 0.005 in 253 of 259 (run, method) pairs. The exceptions are five clustered-conformal runs, short by up to 0.012, whose rare groups receive the pooled threshold as in the reference implementation, and one shrinkage run (0.005).

B.11. Real data: U_3 and paired comparisons (post hoc)

Computed from the logged per-group counts of the real-data runs, without reading the data again. Both procedures were designed after the real-data results were seen, so this analysis is exploratory. Runs `r1_real/20261013T172140Z-9206add` and `summarize_r1/20261013T185458Z-19f1c92`.

Table 16. Design calculator on the real test counts (post hoc). For each primary real-data cell, the test counts of split r define simulated test set r (200 per cell) for a method with exact coverage τ in every group; entries are the share of test sets with naive $\mathrm{CovGap}_{\mathcal{G}_2} > U$. G_2: mean number of groups with $m_g \geq 2$; small: share of groups with $m_g < 20$. U_1 certifies in none, and with heterogeneous coverage (s.d. 0.03) neither U_1 nor U_3 does. Monte Carlo s.e. at most 0.026. Runs `design_profiles/20261013T183303Z-92a4797` and `summarize_r2/20261013T184553Z-4cd2cb1`.

				$\alpha = 0.10$		$\alpha = 0.05$	
Dataset (groups)	Score	G_2	small	U_0 (not valid)	U_3	U_0 (not valid)	U_3
ACS (state×race)	CQR	393	0.35	0.48	0.00	0.28	0.00
Airlines (origin)	CQR	334	0.27	0.33	0.00	0.18	0.00
Airlines (origin)	abs	334	0.27	0.28	0.00	0.18	0.00
Zurich (stop)	CQR	1525	0.06	0.88	0.00	0.92	0.00
Zurich (stop)	abs	1526	0.06	0.91	0.00	0.93	0.00
DBpedia (L3 class)	APS	219	0.00	0.96	0.01	0.96	0.01
DBpedia (L3 class)	LAC	219	0.00	0.97	0.01	0.96	0.01
CIFAR-100 (class)	APS	100	0.00	0.99	0.17	0.99	0.06
CIFAR-100 (class)	LAC	100	0.00	0.99	0.16	1.00	0.04
Diamonds (color×clarity)	CQR	56	0.10	0.10	0.00	0.11	0.00
Diamonds (color×clarity)	abs	56	0.10	0.07	0.01	0.10	0.01
House sales (zipcode)	CQR	70	0.08	0.68	0.01	0.66	0.01
House sales (zipcode)	abs	70	0.09	0.69	0.00	0.66	0.00

Table 17. As Table 1 but on the coarser grid $25 \cdot 2^k$ (smallest grid value reaching 80% of 200 replications), with heterogeneous coverage (true RMS about 0.03). Run `design_rule/20261013T172142Z-9206add`.

	$\tau = 0.90$			$\tau = 0.95$		
m	U_0	U_1	U_3	U_0	U_1	U_3
2	200	800	800	800	3,200	3,200
3	50	800	400	400	1,600	1,600
5	50	400	400	100	1,600	800
10	100	800	400	100	1,600	400
20	100	1,600	400	200	3,200	1,600
50	200	3,200	800	800	12,800	3,200
100	800	12,800	3,200	$> 102{,}400$	$> 102{,}400$	$> 102{,}400$

B.12. Further point-estimation comparators (post hoc)

Added after the synthetic study had been analysed, so the comparison is descriptive. g-modeling (Efron, 2016) fits a log-spline prior for c_g (natural spline, 5 degrees of freedom, penalty $c_0 = 1$) by penalized marginal likelihood, stratified by m_g as our estimator is, and plugs the fitted prior into CovGap and SF. It was run on the draws of the first synthetic study regenerated from their seeds (replications 0–199 of all 1344 cells); the regenerated naive CovGap matched the logged one in all 268,800 rows. The LP midpoint is the midpoint of the moment-LP bounds for CovGap, from the logged rows. Against the exact truth, the median over cells of our mean absolute error over g-modeling's is 0.32 for CovGap (ours lower in 89% of cells; 0.24 outside and 0.45 in the many-small regime) and 0.38 for SF (88%); over the LP midpoint's it is 0.17 (98%). With these default settings g-modeling's error is a median 1.37 times that of the better of the naive and beta-binomial plug-ins. Runs `r1_gmodel/20261013T172723Z-db14078` and `summarize_r1/20261013T185458Z-19f1c92`.

Table 18. All real-data runs. Columns as in Table 3.

Dataset (groups)	Score	Model	α	$\lvert\mathcal{G}_1\rvert$	small	noise	naive / est.	floor / naive	U_1 / naive RMS	r_{K}
ACS (state×race)	CQR	LightGBM	0.10	408	0.35	1.01	3.20	0.96	1.60	0.90
ACS (state×race)	CQR	LightGBM	0.05	408	0.35	1.01	2.85	1.02	2.30	1.00
ACS (state×race)	norm	ridge	0.10	408	0.35	0.87	2.98	0.92	1.51	1.00
Airlines (origin)	CQR	LightGBM	0.10	352	0.27	0.61	1.61	0.65	1.38	1.00
Airlines (origin)	CQR	LightGBM	0.05	352	0.27	0.86	1.61	0.58	1.69	0.43
Airlines (origin)	norm	ridge	0.10	352	0.27	0.66	1.59	0.59	1.36	0.62
Airlines (origin)	abs	LightGBM	0.10	352	0.27	0.88	1.87	0.76	1.65	0.62
Airlines (origin)	abs	LightGBM	0.05	352	0.27	1.07	2.66	0.88	2.50	0.90
Airlines (origin)	abs	ridge	0.10	352	0.27	0.80	2.03	0.77	1.68	1.00
Zurich (stop)	CQR	LightGBM	0.10	1528	0.06	0.41	1.25	0.54	1.17	1.00
Zurich (stop)	CQR	LightGBM	0.05	1528	0.06	0.38	1.27	0.60	1.44	0.90
Zurich (stop)	norm	ridge	0.10	1528	0.06	0.16	1.08	0.40	1.14	0.90
Zurich (stop)	abs	LightGBM	0.10	1528	0.06	0.22	1.15	0.44	1.16	1.00
Zurich (stop)	abs	LightGBM	0.05	1528	0.06	0.21	1.19	0.49	1.34	0.90
Zurich (stop)	abs	ridge	0.10	1528	0.06	0.20	1.13	0.41	1.14	1.00
DBpedia (L3 class)	APS	TF-IDF LR	0.10	219	0.00	0.29	1.27	0.57	2.45	1.00
DBpedia (L3 class)	APS	TF-IDF LR	0.05	219	0.00	0.34	1.28	0.61	3.72	0.90
DBpedia (L3 class)	LAC	TF-IDF LR	0.10	219	0.00	0.05	1.03	0.20	1.23	1.00
DBpedia (L3 class)	LAC	TF-IDF LR	0.05	219	0.00	0.06	1.05	0.24	1.49	1.00
CIFAR-100 (class)	APS	ViT-S probe	0.10	100	0.00	0.37	1.32	0.64	2.31	0.90
CIFAR-100 (class)	APS	ViT-S probe	0.05	100	0.00	0.40	1.36	0.67	3.31	0.90
CIFAR-100 (class)	APS	ResNet-50 probe	0.10	100	0.00	0.35	1.18	0.59	2.26	1.00
CIFAR-100 (class)	LAC	ViT-S probe	0.10	100	0.00	0.30	1.21	0.57	2.17	0.71
CIFAR-100 (class)	LAC	ViT-S probe	0.05	100	0.00	0.38	1.28	0.64	3.19	0.90
CIFAR-100 (class)	LAC	ResNet-50 probe	0.10	100	0.00	0.34	1.20	0.59	2.22	0.90
Diamonds (color×clarity)	CQR	LightGBM	0.10	56	0.10	0.44	1.34	0.62	2.89	0.90
Diamonds (color×clarity)	CQR	LightGBM	0.05	56	0.10	0.42	1.19	0.51	3.31	0.62
Diamonds (color×clarity)	norm	ridge	0.10	56	0.10	0.12	1.14	0.42	2.03	0.90
Diamonds (color×clarity)	abs	LightGBM	0.10	56	0.10	0.26	1.11	0.42	2.05	0.90
Diamonds (color×clarity)	abs	LightGBM	0.05	56	0.10	0.23	1.17	0.41	2.34	1.00
Diamonds (color×clarity)	abs	ridge	0.10	56	0.10	0.12	1.06	0.35	1.82	1.00
House sales (zipcode)	CQR	LightGBM	0.10	70	0.08	0.57	1.59	0.77	2.82	0.90
House sales (zipcode)	CQR	LightGBM	0.05	70	0.09	0.62	1.75	0.78	3.71	0.90
House sales (zipcode)	norm	ridge	0.10	70	0.09	0.21	1.20	0.56	1.94	0.81
House sales (zipcode)	abs	LightGBM	0.10	70	0.09	0.55	1.27	0.67	2.67	1.00
House sales (zipcode)	abs	LightGBM	0.05	70	0.09	0.56	1.31	0.66	3.61	0.81
House sales (zipcode)	abs	ridge	0.10	70	0.08	0.15	1.16	0.52	1.77	0.81

Table 19. The three cells of the pre-registered real-data predictions (CQR, primary model, $\alpha = 0.1$), per method: mean over 200 splits of marginal coverage, mean interval length, naive and estimated CovGap, the oracle floor, naive and debiased RMS, and U_1.

Dataset	Method	coverage	length	naive CovGap	est. CovGap	floor	naive RMS	deb. RMS	U_1
ACS	clustered	0.900	2.155	0.053	0.016	0.052	0.082	0.006	0.132
ACS	conditional (ridge)	0.900	2.154	0.053	0.016	0.052	0.082	0.004	0.131
ACS	Mondrian	0.909	2.244	0.084	0.084	0.052	0.090	0.090	0.165
ACS	shrink $\kappa = 10$	0.904	2.223	0.059	0.023	0.052	0.084	0.015	0.133
ACS	shrink $\kappa = 200$	0.903	2.176	0.054	0.017	0.052	0.082	0.005	0.131
ACS	shrink $\kappa = 50$	0.904	2.195	0.056	0.020	0.052	0.083	0.007	0.132
ACS	standard	0.900	2.154	0.053	0.016	0.052	0.082	0.004	0.131
Airlines	clustered	0.899	52.468	0.071	0.043	0.047	0.112	0.069	0.155
Airlines	conditional (ridge)	0.899	52.400	0.071	0.043	0.047	0.112	0.069	0.154
Airlines	Mondrian	0.910	57.412	0.080	0.079	0.047	0.087	0.086	0.164
Airlines	shrink $\kappa = 10$	0.904	55.562	0.078	0.052	0.047	0.114	0.073	0.156
Airlines	shrink $\kappa = 200$	0.903	53.019	0.072	0.045	0.047	0.112	0.070	0.155
Airlines	shrink $\kappa = 50$	0.904	53.930	0.075	0.048	0.047	0.113	0.071	0.155
Airlines	standard	0.900	52.469	0.071	0.042	0.047	0.112	0.069	0.154
House sales	clustered	0.898	0.529	0.048	0.031	0.036	0.065	0.041	0.182
House sales	conditional (ridge)	0.901	0.533	0.046	0.028	0.036	0.063	0.039	0.181
House sales	Mondrian	0.922	0.629	0.063	0.055	0.036	0.075	0.067	0.190
House sales	shrink $\kappa = 10$	0.915	0.594	0.053	0.040	0.036	0.068	0.051	0.184
House sales	shrink $\kappa = 200$	0.905	0.540	0.046	0.027	0.036	0.062	0.039	0.181
House sales	shrink $\kappa = 50$	0.911	0.558	0.046	0.029	0.036	0.062	0.040	0.181
House sales	standard	0.900	0.531	0.047	0.030	0.036	0.064	0.041	0.181

Table 20. Marginal coverage minus $1 - \alpha$, per method over the 37 runs (each a mean over 200 splits), and the number of runs more than 0.005 below nominal.

Method	min	median	max	runs below -0.005
clustered	-0.012	0.000	+0.004	5
conditional (ridge)	-0.001	0.000	+0.003	0
Mondrian	+0.008	+0.020	+0.059	0
shrink $\kappa = 10$	-0.002	+0.006	+0.018	0
shrink $\kappa = 200$	-0.004	+0.003	+0.006	0
shrink $\kappa = 50$	-0.005	+0.005	+0.011	1
standard	-0.001	0.000	+0.001	0

Table 21. Post hoc, primary runs at $\alpha = 0.1$: medians over the 7 methods of U_3 over the naive RMS and over U_1 (means over 200 splits), and the number of the 21 method pairs that the paired test separates (Holm over the pairs) in at least half of the splits. The naive CovGap and the naive RMS exceed U_3 in none of the $51{,}800$ evaluations of the 37 runs.

Dataset (groups)	Score	U_3 / naive RMS	U_3/U_1	pairs separated
ACS (state×race)	CQR	1.31	0.82	13
Airlines (origin)	CQR	1.22	0.88	9
Airlines (origin)	abs	1.36	0.83	10
Zurich (stop)	CQR	1.08	0.92	14
Zurich (stop)	abs	1.11	0.95	17
DBpedia (L3 class)	APS	1.26	0.51	12
DBpedia (L3 class)	LAC	1.08	0.88	15
CIFAR-100 (class)	APS	1.13	0.49	2
CIFAR-100 (class)	LAC	1.13	0.52	1
Diamonds (color×clarity)	CQR	1.73	0.60	3
Diamonds (color×clarity)	abs	1.44	0.70	0
House sales (zipcode)	CQR	1.66	0.58	0
House sales (zipcode)	abs	1.58	0.59	1

Table 22. Post hoc: the runs where the method with the smallest naive CovGap (naive leader) differs from the one with the smallest estimated CovGap (estimated leader). Δ: mean over splits of the paired estimate of $\mathrm{MSE}_{\text{naive leader}} - \mathrm{MSE}_{\text{estimated leader}}$ ($\times 10^3$); shares of splits with $p < 0.05$, without and with Holm over the 21 pairs.

Data	Score	Model	α	naive leader	est. leader	Δ	$p < 0.05$	Holm
ACS	CQR	LightGBM	0.10	standard	conditional (ridge)	-0.01	0.055	0.010
Airlines	CQR	LightGBM	0.05	Mondrian	standard	$+1.13$	0.320	0.045
Airlines	abs	LightGBM	0.10	shrink $\kappa = 200$	conditional (ridge)	-0.08	0.065	0.000
CIFAR-100	LAC	ViT-S probe	0.10	shrink $\kappa = 10$	shrink $\kappa = 50$	-0.28	0.135	0.000
Diamonds	CQR	LightGBM	0.05	Mondrian	conditional (ridge)	-0.50	0.010	0.000
Diamonds	abs	LightGBM	0.10	Mondrian	conditional (ridge)	-3.64	0.005	0.000